Extended supplementary information for "External Evaluation of EEG Dementia Classifiers and Their Referral Thresholds Across Independent Clinical Cohorts"

This document extends the supplementary material submitted with the paper; the main text cites its items by their E numbers. Items numbered S (the task design, training recipe, age and artifact analyses in Supplementary Tables S1–S4; the all-model AUC overview, capacity, and learning curves in Supplementary Figs. S1–S3) are in the submitted supplement and are not repeated here. The TRIPOD+AI checklist is Section E12.

E1. Cohorts, tasks, and acquisition matching

Table E1 gives each cohort's setting, diagnostic criteria, recording system and condition, recording period, and ethics approval as published by its source, together with the age, sex, cognitive score, and education of the retained subjects. A cohort was included if it provided 24 artifact-free 4 s windows within a common 235 s span, could be mapped to the 10–20 montage, and had at least two diagnostic classes. CAUEEG retained 1,098 of 1,120 recordings [1], ds004504 87 of 88 [2], P-ADIC 166 of 167 [3], BrainLat all 80 [4], GENEEG all 63 [5], and MISP 48 of 61 (29 of 32 controls, 19 of 29 MCI; retention by class in Fig. E1a). MCI_Dataset (controls 70 s long; duration alone gave AUC 0.94), AlzEEG (about 8 s per recording), and Mendeley, ADSZ, and APAVA (23–48 subjects) were excluded.

TABLE E1: Cohort provenance and characteristics of the retained subjects. Provenance is transcribed from each source publication or released metadata. Age and cognitive scores are median (interquartile range); female, percentage of subjects with sex recorded (CAUEEG and GENEEG release no sex). P-ADIC MMSE is released for patients only. No cohort releases treatments or comorbidities per subject. AAN, American Academy of Neurology; NIA-AA, National Institute on Aging and Alzheimer's Association; NINCDS-ADRDA, National Institute of Neurological and Communicative Disorders and Stroke and Alzheimer's Disease and Related Disorders Association; MoCA, Montreal Cognitive Assessment.

Class	*n*	Age (years)	Female (%)	Cognitive score	Education (years)
CAUEEG. Memory clinic, Chung-Ang University Hospital, Korea; retrospective clinical EEG. Dementia by NINCDS-ADRDA and DSM-IV criteria; MCI by CDR 0.5 and neuropsychological impairment ($\geq$1 SD below age- and education-adjusted norms); normal cognition on the same battery. Comet AS40 amplifier (Grass Telefactor), 19 ch, 200 Hz; eyes open, eyes closed, and photic stimulation in one recording. Period: Aug 2012–Mar 2020. Ethics: Chung-Ang University Hospital IRB, registration no. 2009-005-19331.					
CN	455	65 (58–72)	–	–	–
MCI	410	75 (69–79)	–	–	–
AD	221	78 (71–81)	–	–	–
FTD	12	75 (70–81)	–	–	–
ds004504. Second Department of Neurology, AHEPA University Hospital, Thessaloniki, Greece. AD and FTD diagnosed by neurologists; MMSE released. Nihon Kohden EEG 2100, 19 ch, 500 Hz; eyes closed. Period: Not stated. Ethics: AHEPA Scientific and Ethics Committee, protocol 142/12-04-2023.					
CN	29	67 (64–71)	38	MMSE 30 (30–30)	–
AD	36	67 (61–71)	67	MMSE 20 (16–20)	–
FTD	22	64 (61–71)	41	MMSE 22 (20–24)	–
P-ADIC. Two Israeli medical centers; routine clinical EEG; controls referred for unrelated indications. AD and MCI by two senior neurologists (NIA-AA criteria); controls with neuropsychiatric exclusions. Nihon Kohden, 19 ch, 500 Hz (26 controls released at 200 Hz); eyes open and closed. Period: 2011–2019. Ethics: Rabin Medical Center IRB 0275-20-RMC; retrospective, consent waived.					
CN	96	54 (42–65)	66	–	–
MCI	28	73 (70–78)	50	MMSE 28 (27–29)	–
AD	42	72 (64–78)	51	MMSE 22 (14–26)	–
BrainLat. Multicenter Latin-American network; retained subjects from Argentina and Chile. Probable AD (NINCDS-ADRDA); probable bvFTD (revised criteria); MoCA released. Biosemi Active Two, 128 ch, 512 Hz, linked mastoids; eyes closed, 10 min. Period: Not stated. Ethics: Ethics boards of each site; written informed consent including data sharing.					
CN	32	72 (64–75)	62	MoCA 26 (25–28)	16.5 (14–18)
AD	30	76 (70–81)	60	MoCA 17 (15–20)	8.5 (6–12)
FTD	18	66 (60–79)	29	MoCA 23 (19–26)	16.0 (14–18)
GENEEG. General-practice clinics. MCI by clinicians following the 2018 AAN guideline. 19-ch system with T5 and T6 excluded (17 ch), 250 Hz; eyes closed during an auditory oddball protocol. Period: Not stated. Ethics: Johns Hopkins University IRB00348520.					
CN	35	–	–	–	–

Class	n	Age (years)	Female (%)	Cognitive score	Education (years)
MCI	28	–	–	–	–

MISP. Cardiac catheterization units of Sina and Nour hospitals, Isfahan, Iran; coronary angiography within the past year; aged 60–77 (set 1); elementary education or higher; released as set 1 (16 CN, 11 MCI) and set 2 (16 CN, 18 MCI), analyzed last. MCI by Petersen criteria with MMSE 21–26, confirmed with NUCOG; controls MMSE >26; dementia, major psychiatric disorder, substance misuse, and head trauma excluded. Galileo NT (EBNeuro), 19 ch, 256 Hz, impedance <5 kΩ; eyes closed, 30 min, morning, vigilance monitored. Period: Not stated. Ethics: Research deputy, Isfahan University of Medical Sciences; written informed consent.

CN	29	–	–	MMSE >26 (criterion)	–
MCI	19	–	–	MMSE 21–26 (criterion)	–

Each task is a separate binary model with cognitively normal (CN) participants as the negative class (Supplementary Table S1). P-ADIC has no FTD, so its dementia task uses AD and its any-impairment task AD plus MCI; its controls are restricted to ages 55 and over because they range from 19 to 80 years while its patients are 55 to 87. ds004504 and BrainLat have no MCI. BrainLat releases its controls (`rs_eeg` files) and patients (`rs-HEP_eeg` files) as two exports whose amplitude differs by a near-constant factor (3.5–4.3 at every frequency from 1 to 40 Hz). Its retained subjects come from Argentina (19 controls, 15 AD, 12 FTD), Chile (13 controls, 15 AD), a later release (5 FTD, site not given), and one FTD recording absent from the released demographics file; the country analyses therefore use 42 of the 48 dementia cases. Robust scaling (Eq. 1) removes a constant gain exactly, and no BrainLat window exceeded the 200 μV rejection threshold, so the gain difference does not reach any feature; Fig. E1b gives the acquisition checks. BrainLat's 128 electrodes were reduced to the 10–20 set by taking, for each 10–20 label, the Biosemi electrode nearest in three-dimensional position, before common-average referencing over the 19.

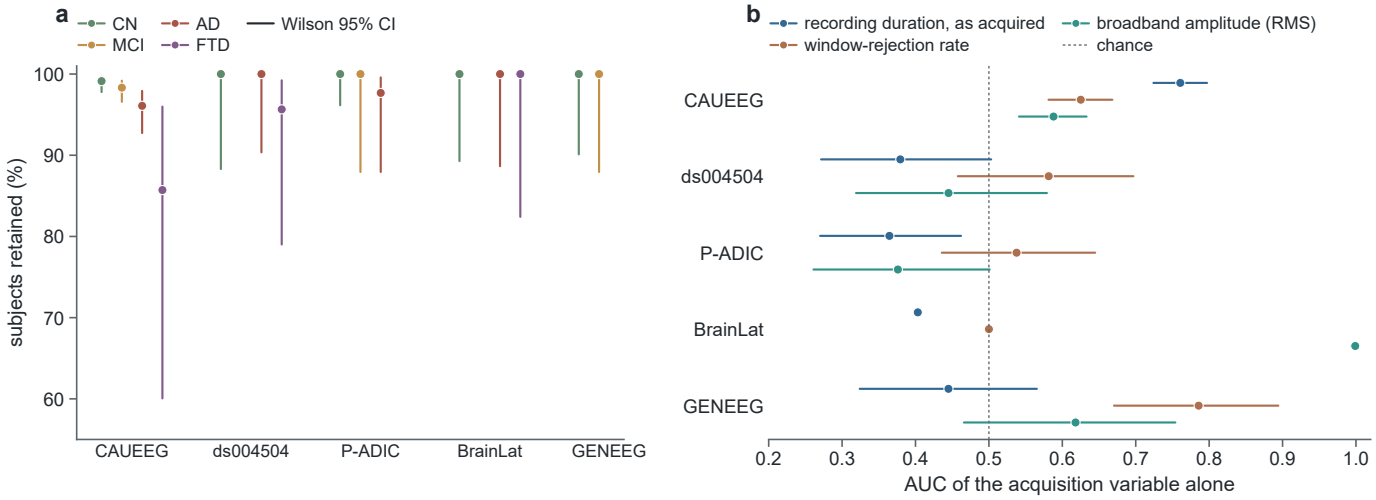


Fig. E1. Retention and acquisition checks. (a) Percentage of offered subjects retained after the 24-window quota, by class, with Wilson 95% intervals. (b) AUC of recording duration as acquired (before truncation), of the window-rejection rate, and of broadband amplitude (root-mean-square voltage of the matched windows before robust scaling), each used alone to separate controls from dementia (from MCI at GENEEG; AD at BrainLat), with subject-bootstrap 95% intervals (2,000 resamples; none for the BrainLat duration and rejection AUCs, which are degenerate). The protocol equalizes duration; amplitude separates the classes only at BrainLat, where it reflects the export gain. MISP, analyzed last, is reported in Table I of the main text and Section E4.

E2. Features and Model

ICA (main text, Section II-B) was fitted on up to 300 s of each recording high-passed at 1 Hz, with the component count set by the data rank. Let i index a subject, j one of its $K = 24$ windows, c an electrode, and t one of the $T = 1024$ samples of a window (sampling rate $f_s = 256$ Hz); $x_{ijc}[t]$ is the signal. Each channel is centered by its window median and divided by its interquartile range,

$$\tilde{x}_{ijc}[t] = \frac{x_{ijc}[t] - \mathrm{median}_t\, x_{ijc}[t]}{\mathrm{IQR}_t(x_{ijc}[t])}, \tag{1}$$

set to zero when the denominator is zero. A one-sided Welch PSD $S_{ijc}(f)$ (2 s Hann segments, 50% overlap, 257 bins) gives relative band power $P_{ijc}(b) = \sum_{f\in b} S_{ijc}(f)/Z_{ijc}$ with $Z_{ijc} = \sum_f S_{ijc}(f) + 10^{-10}$ for δ, θ, α, β, γ ($[0.5,4)$, $[4,8)$, $[8,13)$, $[13,30)$, $[30,45)$ Hz). The individual alpha frequency q_{ijc} is the parabolically interpolated maximum of $\log S$ over $6 \le f < 14$ Hz, its prominence ψ_{ijc} the peak minus the median log PSD on that interval, and the edge e_{ijc} the lowest frequency in $[1,30)$ Hz at which cumulative power reaches 95%. The aperiodic fit $\log_{10} S_{ijc}(f) = \kappa_{ijc} - \chi_{ijc} \log_{10} f + \eta_{ijc}(f)$ uses $2 \le f < 40$ Hz excluding $7 \le f < 14$ Hz, where $\eta_{ijc}(f)$ is the fit residual, giving the exponent χ and offset κ. The log-ratios are $R^{TA} = \log(1 + P(\theta)/P(\alpha))$ and $R^{slow} = \log(1 + (P(\delta) + P(\theta))/(P(\alpha) + P(\beta)))$, and spectral entropy is the normalized Shannon entropy of S/Z. Fixed

constants scale q as $(q-10)/3$, ψ as $\psi/3$, e as $(e-20)/10$, χ as $\chi-1.5$, and each log-ratio by $1/2$. The 16-feature variant adds θ and α relative power in $[q-4, q-2)$ and $[q-2, q+2)$ Hz, the offset $\kappa+12$, and the log alpha-power ratio to the homologous electrode (midline electrodes map to themselves). No constant is fitted to any receiving-site recording. The 123-dimensional QEEG vector of the classical baselines holds 95 channel-level and 25 regional band powers and three whole-head summaries, reduced to the subject median.

SpectraLite's feature encoder maps the 12 features of an electrode through a linear layer to 128 units, LayerNorm, GELU, dropout 0.3, and a second linear layer; the position encoder applies the same structure to 51 inputs: the electrode's unit-sphere coordinate $\mathbf{p}$ and $\sin(2^l\mathbf{p})$ and $\cos(2^l\mathbf{p})$ for $l = 0, \ldots, 7$. Their sum is pooled by masked gated attention (main text, Eq. 1), and the head applies LayerNorm, a 128-unit linear layer with GELU and dropout, and a two-class linear output. SpectraLite has 18,432 parameters in the feature encoder, 23,424 in the position encoder, 16,577 in the pooling, and 17,026 in the head (75,459); the 16-feature variant has 512 more in the feature encoder (75,971).

NeuroChronoGraph. NeuroChronoGraph (NCG) is the graph–temporal family developed in this project (code in the repository). In NCG-v5 (915,401 trained of 964,000 declared parameters) each electrode node carries SpectraLite's 12 spectral features and position embedding plus a convolutional encoding of that electrode's raw window. An adaptive layer builds the adjacency from the similarity of node embeddings, two gated graph convolutions update the nodes, and gated attention pools them; a lobe-coupling module summarizes interactions between lobes, and a two-layer temporal encoder reads the multichannel waveform. The three streams are concatenated and classified. NCG-v2 (6.0 M declared) and NCG-v3 (6.3 M declared) are earlier versions that share one trunk: a convolutional encoder of each electrode's raw window with learned electrode embeddings (128-dimensional nodes); an adaptive layer that builds the adjacency from attention between nodes; three gated graph convolutions; a modular transformer that attends within and between five lobe modules (frontal, central, temporal, parietal, occipital); and a four-layer, eight-head temporal transformer on a multichannel convolutional encoding. The lobe-level output, the temporal output, and the mean node embedding are concatenated (512 dimensions) and classified. NCG-v2 ends in a hierarchical screening, staging, and subtype head, of which only the screening output was trained here; NCG-v3 ends in one two-layer classifier. Both also carry band-specific encoders with cross-band attention, MMSE and evidential-uncertainty heads, and clinical-metadata conditioning, none of which received input or loss here: every version scored single 4 s windows from the EEG alone with a cross-entropy loss on the class output, and v5's phase-coupling prior and band-feature inputs were not supplied either. The learned adjacency therefore reflects similarity of node features, not measured connectivity. The paper's parameter counts are those that receive a gradient from the classification loss on an EEG-only batch, counted from the instantiated models (experiment 227): 5,382,215 for NCG-v2 and NCG-v3 alike, 915,401 for NCG-v5, and every declared parameter for the other networks. Counting declared parameters instead changes every capacity correlation in Supplementary Fig. S2 by at most 0.01. The NCG ladder adds NCG-v5's raw-waveform, graph, and lobe streams to SpectraLite one at a time (+raw; +graph; +raw+graph; +raw+graph+lobe) with the same recipe and data, so it isolates what each stream adds; SpectraLite is the ladder's first rung. Networks trained from scratch used the recipe in Supplementary Table S2; the fine-tuned encoders and probes are described in Section E3.

E3. Benchmark

Supplementary Fig. S1 gives every model at every task and cohort. Table E2 gives the primary task in full (seed standard deviations, paired differences from the QEEG forest, equivalence), and Fig. E2 shows those differences against the equivalence margins at all three external cohorts. Every network trained from scratch, and both fine-tuned encoders, have three training seeds and are reported as seed ensembles; the forests use three seeds; gradient boosting, the support vector machine, logistic regression, and the frozen probes are deterministic fits.

Table E3 gives the paired difference of every model from SpectraLite for the dementia and AD tasks at each external cohort; none survived Holm correction.

TABLE E3: Paired comparison of every model with SpectraLite at each external cohort. Δ = model − SpectraLite AUC (three-seed ensembles where available) with a paired subject-bootstrap 95% interval (4,000 resamples, stratified by class); P_{Holm}, Holm-adjusted over every comparator in the task-cohort family. The 16-feature variant is exploratory (Section E3).

Model	ds004504		P-ADIC		BrainLat	
	Δ [95% CI]	P_{Holm}	Δ [95% CI]	P_{Holm}	Δ [95% CI]	P_{Holm}
Control versus dementia						
EEGPT (frozen probe)	+0.034 [−0.047, +0.116]	1.00	+0.002 [−0.080, +0.083]	1.00	+0.022 [−0.088, +0.139]	1.00
16-feature variant	+0.029 [−0.002, +0.067]	1.00	+0.011 [−0.026, +0.050]	1.00	+0.042 [−0.002, +0.092]	1.00
LaBraM (fine-tuned)	+0.026 [−0.028, +0.082]	1.00	+0.012 [−0.045, +0.071]	1.00	+0.027 [−0.066, +0.118]	1.00
LEAD (frozen probe)	+0.020 [−0.026, +0.066]	1.00	–	–		
ATCNet	+0.020 [−0.045, +0.087]	1.00	+0.014 [−0.060, +0.088]	1.00	+0.078 [−0.010, +0.171]	1.00
CBraMod (frozen probe)	+0.017 [−0.046, +0.080]	1.00	−0.028 [−0.099, +0.040]	1.00	+0.070 [−0.017, +0.162]	1.00
NCG-v3	+0.017 [−0.025, +0.060]	1.00	+0.033 [−0.019, +0.088]	1.00	−0.010 [−0.091, +0.070]	1.00
NCG-v2	+0.015 [−0.027, +0.062]	1.00	+0.042 [−0.005, +0.091]	1.00	−0.013 [−0.089, +0.062]	1.00
NCG ladder +raw	+0.012 [−0.010, +0.035]	1.00	−0.007 [−0.039, +0.024]	1.00	+0.020 [−0.025, +0.068]	1.00

Model	ds004504		P-ADIC		BrainLat	
NCG ladder +raw+graph	+0.010 [−0.018, +0.038]	1.00	+0.002 [−0.052, +0.053]	1.00	+0.016 [−0.059, +0.090]	1.00
LaBraM (frozen probe)	+0.009 [−0.057, +0.075]	1.00	−0.053 [−0.124, +0.020]	1.00	+0.060 [−0.063, +0.188]	1.00
NCG ladder +raw+graph+lobe	+0.005 [−0.026, +0.036]	1.00	+0.000 [−0.050, +0.049]	1.00	−0.003 [−0.081, +0.074]	1.00
CBraMod (fine-tuned)	+0.004 [−0.040, +0.047]	1.00	+0.027 [−0.027, +0.082]	1.00	−0.090 [−0.194, +0.006]	1.00
NCG ladder +graph	+0.002 [−0.018, +0.022]	1.00	+0.014 [−0.008, +0.037]	1.00	+0.001 [−0.030, +0.033]	1.00
EEG-ITNet	−0.005 [−0.087, +0.073]	1.00	−0.029 [−0.109, +0.045]	1.00	+0.101 [+0.006, +0.199]	1.00
EEGNetv4	−0.006 [−0.086, +0.073]	1.00	−0.003 [−0.090, +0.083]	1.00	+0.100 [+0.005, +0.196]	1.00
Forest (QEEG-123)	−0.006 [−0.067, +0.054]	1.00	+0.021 [−0.052, +0.094]	1.00	−0.025 [−0.117, +0.067]	1.00
EEGNeX	−0.007 [−0.075, +0.062]	1.00	−0.007 [−0.088, +0.072]	1.00	+0.042 [−0.055, +0.143]	1.00
Forest (node features)	−0.010 [−0.065, +0.045]	1.00	+0.024 [−0.045, +0.096]	1.00	+0.004 [−0.071, +0.083]	1.00
SVM	−0.016 [−0.086, +0.052]	1.00	+0.008 [−0.070, +0.089]	1.00	−0.012 [−0.110, +0.089]	1.00
EEGConformer	−0.020 [−0.097, +0.059]	1.00	−0.035 [−0.136, +0.066]	1.00	+0.005 [−0.113, +0.120]	1.00
Gradient boosting	−0.022 [−0.090, +0.043]	1.00	−0.033 [−0.107, +0.039]	1.00	+0.029 [−0.080, +0.138]	1.00
NCG-v5	−0.024 [−0.066, +0.016]	1.00	−0.008 [−0.060, +0.044]	1.00	−0.004 [−0.072, +0.066]	1.00
Deep4Net	−0.037 [−0.136, +0.065]	1.00	−0.037 [−0.116, +0.040]	1.00	−0.012 [−0.131, +0.107]	1.00
ShallowFBCSPNet	−0.042 [−0.130, +0.041]	1.00	−0.046 [−0.139, +0.052]	1.00	+0.010 [−0.105, +0.125]	1.00
Logistic regression	−0.080 [−0.173, +0.012]	1.00	−0.018 [−0.129, +0.099]	1.00	−0.058 [−0.181, +0.066]	1.00
Control versus AD						
16-feature variant	+0.012 [−0.012, +0.042]	1.00	+0.004 [−0.027, +0.034]	1.00	+0.041 [−0.003, +0.093]	1.00
NCG ladder +graph	+0.002 [−0.014, +0.018]	1.00	+0.008 [−0.016, +0.033]	1.00	−0.004 [−0.039, +0.031]	1.00
NCG ladder +raw+graph+lobe	−0.005 [−0.031, +0.019]	1.00	−0.002 [−0.046, +0.043]	1.00	+0.031 [−0.045, +0.110]	1.00
NCG ladder +raw+graph	−0.008 [−0.034, +0.018]	1.00	+0.002 [−0.049, +0.052]	1.00	+0.029 [−0.042, +0.103]	1.00
NCG ladder +raw	−0.010 [−0.039, +0.018]	1.00	−0.021 [−0.053, +0.007]	1.00	+0.023 [−0.020, +0.070]	1.00
Forest (node features)	−0.011 [−0.054, +0.031]	1.00	+0.012 [−0.056, +0.083]	1.00	+0.024 [−0.051, +0.100]	1.00
LaBraM (fine-tuned)	−0.011 [−0.062, +0.039]	1.00	−0.005 [−0.061, +0.050]	1.00	+0.051 [−0.037, +0.146]	1.00
CBraMod (frozen probe)	−0.013 [−0.074, +0.047]	1.00	−0.031 [−0.101, +0.039]	1.00	+0.080 [−0.012, +0.182]	1.00
LEAD (frozen probe)	−0.015 [−0.068, +0.035]	1.00	–	–		
ATCNet	−0.017 [−0.080, +0.046]	1.00	+0.009 [−0.062, +0.076]	1.00	+0.095 [+0.003, +0.200]	1.00
NCG-v3	−0.019 [−0.060, +0.017]	1.00	+0.024 [−0.028, +0.076]	1.00	+0.016 [−0.060, +0.095]	1.00
EEGPT (frozen probe)	−0.023 [−0.094, +0.046]	1.00	−0.022 [−0.105, +0.053]	1.00	+0.032 [−0.084, +0.149]	1.00
EEGNeX	−0.027 [−0.097, +0.042]	1.00	−0.009 [−0.091, +0.070]	1.00	+0.099 [−0.005, +0.213]	1.00
CBraMod (fine-tuned)	−0.030 [−0.074, +0.010]	1.00	+0.022 [−0.034, +0.078]	1.00	−0.026 [−0.127, +0.066]	1.00
Forest (QEEG-123)	−0.031 [−0.096, +0.025]	1.00	+0.021 [−0.054, +0.095]	1.00	+0.015 [−0.076, +0.111]	1.00
NCG-v2	−0.032 [−0.075, +0.006]	1.00	+0.023 [−0.025, +0.072]	1.00	+0.017 [−0.056, +0.094]	1.00
EEG-ITNet	−0.037 [−0.116, +0.037]	1.00	−0.022 [−0.091, +0.046]	1.00	+0.128 [+0.033, +0.236]	0.14
NCG-v5	−0.037 [−0.083, −0.001]	1.00	−0.005 [−0.053, +0.044]	1.00	−0.002 [−0.085, +0.080]	1.00
EEGNetv4	−0.041 [−0.123, +0.034]	1.00	−0.033 [−0.117, +0.047]	1.00	+0.117 [+0.015, +0.228]	0.58
SVM	−0.043 [−0.123, +0.028]	1.00	+0.003 [−0.067, +0.076]	1.00	+0.000 [−0.092, +0.093]	1.00
EEGConformer	−0.048 [−0.127, +0.024]	1.00	−0.031 [−0.129, +0.070]	1.00	+0.035 [−0.085, +0.161]	1.00
ShallowFBCSPNet	−0.053 [−0.138, +0.025]	1.00	−0.052 [−0.142, +0.040]	1.00	+0.034 [−0.081, +0.155]	1.00
Gradient boosting	−0.057 [−0.128, +0.005]	1.00	−0.014 [−0.093, +0.059]	1.00	+0.043 [−0.074, +0.161]	1.00
LaBraM (frozen probe)	−0.057 [−0.122, −0.004]	0.97	−0.056 [−0.127, +0.013]	1.00	+0.055 [−0.070, +0.179]	1.00
Deep4Net	−0.078 [−0.174, +0.012]	1.00	−0.047 [−0.122, +0.029]	1.00	+0.019 [−0.105, +0.147]	1.00
Logistic regression	−0.099 [−0.204, −0.005]	0.97	−0.026 [−0.133, +0.084]	1.00	−0.064 [−0.209, +0.079]	1.00

The 16-feature variant. SpectraLite (12 features) was fixed before any external scoring. Five configurations were then trained on CAUEEG and scored for the dementia and AD tasks at ds004504, P-ADIC, and BrainLat: the 12-feature base, feature-space domain randomization, test-time feature re-centering, the 16-feature set, and the 16 features with randomization. The four added descriptors were motivated by healthy-control feature shifts between cohorts (Fig. E3a), and the 16-feature set was adopted because it raised external dementia and AD AUC at the lowest internal cost. Its external values are therefore selection-informed, and the variant is exploratory in every table and figure of this supplement. Table E4 compares the variant with SpectraLite on every task. The 16-feature set lowered internal AUC on the endpoints used for the choice and raised external AUC on them (dementia, pooled +0.026, $P = 0.019$); on the any-impairment and MCI tasks, which were not used, the external difference was near zero. Choosing between the two with each cohort held out selected 16 features every time, but because the candidates themselves were designed after all three cohorts had been seen, this does not make the 16-feature results independent. The prespecified internal criterion of Section E8 selected 12 features for every task.

Capacity checks. Learning curves (Supplementary Fig. S3a) show every model family gaining more internally than externally as the training set grows. Retraining three networks that span the size range under six learning-rate and stopping configurations (Supplementary Fig. S3b) changed ds004504 AUC by at most 0.021 from the fixed recipe; choosing by inner-validation AUC changed it by $+0.013$, $+0.011$, and 0.000. Table E12 gives the capacity correlations with bootstrap intervals, without the NCG versions, and with the 16-feature variant in place of SpectraLite.

Pretrained encoders. Each encoder saw the same 24 matched windows as every other model, in microvolts, common-average referenced, and resampled to its native rate (LaBraM and CBraMod 200 Hz, EEGPT 250 Hz, LEAD 100 Hz), with channel

TABLE E2

DISCRIMINATION FOR CN VS DEMENTIA: SEED ENSEMBLE, WITH THE PER-SEED STANDARD DEVIATION; INTERNAL VALUES ARE CAUEEG OUT-OF-FOLD. SPECTRALITE IS IN BOLD; THE 16-FEATURE VARIANT WAS CHOSEN AFTER EXTERNAL SCORING, SO ITS EXTERNAL VALUES ARE SELECTION-INFORMED. HOLM CORRECTION IS OVER EVERY COMPARATOR IN THE TASK-COHORT FAMILY. PARAMS COUNTS THE PARAMETERS TRAINED BY THE CLASSIFICATION LOSS. LEAD'S INTERNAL VALUE IS NOT INDEPENDENT OF ITS PRETRAINING, WHICH INCLUDED CAUEEG. Δ IS THE PAIRED SUBJECT-BOOTSTRAP DIFFERENCE AT DS004504 FROM THE REFERENCE FOREST ON THE 123-FEATURE VECTOR, HOLM-CORRECTED; THE LAST COLUMN MARKS EQUIVALENCE TO IT WITHIN 0.05 / 0.075 / 0.10 AUC BY TWO ONE-SIDED TESTS.

Model	params	internal	ds004504	P-ADIC	BrainLat	Δ vs forest [95% CI]	p_{Holm}	equiv.
EEGPT (frozen probe)	–	0.842	0.880	0.686	0.635	+0.040 [−0.033, +0.116]	1.000	–/–/–
16-feature variant	76 k	0.837±0.003	0.875±0.004	0.695±0.007	0.656±0.009	+0.035 [−0.028, +0.101]	1.000	–/–/✓
LaBraM (fine-tuned)	–	0.885±0.001	0.872±0.015	0.696±0.004	0.640±0.013	+0.032 [−0.046, +0.109]	1.000	–/–/✓
LEAD (frozen probe)	–	0.853	0.866	–	–	+0.026 [−0.040, +0.093]	1.000	–/–/✓
ATCNet	113 k	0.870±0.005	0.866±0.007	0.698±0.014	0.691±0.013	+0.026 [−0.046, +0.101]	1.000	–/–/✓
CBraMod (frozen probe)	–	0.798	0.863	0.655	0.683	+0.023 [−0.044, +0.091]	1.000	–/–/✓
NCG-v3	5382 k	0.906±0.006	0.863±0.005	0.717±0.007	0.603±0.002	+0.023 [−0.046, +0.093]	1.000	–/–/✓
NCG-v2	5382 k	0.901±0.004	0.861±0.003	0.726±0.011	0.600±0.016	+0.021 [−0.050, +0.092]	1.000	–/–/✓
NCG ladder +raw	103 k	0.877±0.002	0.858±0.002	0.677±0.008	0.633±0.006	+0.018 [−0.046, +0.080]	1.000	–/✓/✓
NCG ladder +raw+graph	442 k	0.895±0.005	0.856±0.008	0.685±0.006	0.629±0.013	+0.016 [−0.047, +0.081]	1.000	–/✓/✓
LaBraM (frozen probe)	–	0.819	0.855	0.630	0.673	+0.015 [−0.054, +0.083]	1.000	–/✓/✓
NCG ladder +raw+graph+lobe	542 k	0.893±0.003	0.851±0.008	0.684±0.008	0.611±0.009	+0.011 [−0.056, +0.077]	1.000	–/✓/✓
CBraMod (fine-tuned)	–	0.864±0.005	0.850±0.003	0.711±0.005	0.523±0.003	+0.010 [−0.060, +0.080]	1.000	–/✓/✓
NCG ladder +graph	414 k	0.860±0.004	0.848±0.006	0.698±0.007	0.614±0.011	+0.008 [−0.054, +0.070]	1.000	–/✓/✓
SpectraLite	75 k	0.856±0.003	0.846±0.014	0.684±0.009	0.613±0.019	+0.006 [−0.054, +0.067]	1.000	–/✓/✓
EEG-ITNet	6 k	0.834±0.001	0.841±0.026	0.655±0.012	0.714±0.024	+0.001 [−0.082, +0.086]	1.000	–/✓/✓
EEGNetv4	2 k	0.840±0.002	0.840±0.014	0.681±0.013	0.714±0.008	+0.000 [−0.075, +0.077]	1.000	–/✓/✓
Forest (QEEG-123)	–	0.810±0.004	0.840±0.008	0.705±0.003	0.588±0.004	reference		
EEGNeX	56 k	0.872±0.008	0.839±0.009	0.677±0.003	0.656±0.009	−0.001 [−0.078, +0.078]	1.000	–/✓/✓
Forest (node features)	–	0.818±0.002	0.836±0.006	0.708±0.003	0.617±0.003	−0.004 [−0.053, +0.043]	1.000	✓/✓/✓
SVM	–	0.819	0.830	0.692	0.601	−0.010 [−0.076, +0.056]	1.000	–/✓/✓
EEGConformer	795 k	0.856±0.005	0.826±0.007	0.649±0.007	0.618±0.002	−0.014 [−0.098, +0.070]	1.000	–/–/✓
Gradient boosting	–	0.794	0.824	0.651	0.642	−0.016 [−0.067, +0.033]	1.000	–/✓/✓
NCG-v5	915 k	0.874±0.001	0.822±0.005	0.676±0.004	0.609±0.001	−0.018 [−0.092, +0.056]	1.000	–/–/✓
Deep4Net	279 k	0.820±0.039	0.809±0.005	0.647±0.006	0.601±0.008	−0.032 [−0.135, +0.078]	1.000	–/–/–
ShallowFBCSPNet	36 k	0.845±0.005	0.804±0.004	0.638±0.001	0.624±0.002	−0.036 [−0.130, +0.056]	1.000	–/–/–
Logistic regression	–	0.784	0.766	0.666	0.555	−0.074 [−0.175, +0.020]	1.000	–/–/–

TABLE E4

THE 16-FEATURE VARIANT AGAINST SPECTRALITE (THREE-SEED ENSEMBLES). Δ = VARIANT − SPECTRALITE WITH PAIRED DELONG 95% INTERVAL AND TWO-SIDED *P* (UNADJUSTED); POOLED EXTERNAL = FIXED-EFFECT INVERSE-VARIANCE MEAN OVER THE EXTERNAL COHORTS. THE 16-FEATURE SET WAS CHOSEN ON EXTERNAL DEMENTIA AND AD AUC AT DS004504, P-ADIC AND BRAINLAT, SO THOSE EXTERNAL ROWS ARE SELECTION-INFORMED; THE INTERNAL ROWS, THE ANY-IMPAIRMENT AND MCI TASKS, AND MISP (ANALYZED LAST) WERE NOT USED FOR THE CHOICE. REPLAYING THE CHOICE WITH EACH COHORT HELD OUT IN TURN SELECTED 16 FEATURES EVERY TIME, FOR BOTH DEMENTIA AND AD.

Task	Evaluation set	*n*	SpectraLite	variant	Δ [95% CI]	*P*
dementia	CAUEEG (internal)	688	0.856	0.837	−0.019 [−0.030, −0.008]	<0.001
	ds004504	87	0.846	0.875	+0.029 [−0.004, +0.062]	0.084
	P-ADIC	88	0.684	0.695	+0.011 [−0.024, +0.047]	0.532
	BrainLat	80	0.613	0.656	+0.042 [−0.003, +0.088]	0.070
	pooled external				+0.026 [+0.004, +0.047]	0.019
AD	CAUEEG (internal)	676	0.876	0.856	−0.020 [−0.029, −0.010]	<0.001
	ds004504	65	0.897	0.909	+0.012 [−0.012, +0.037]	0.314
	P-ADIC	88	0.693	0.697	+0.004 [−0.025, +0.033]	0.780
	BrainLat	62	0.665	0.705	+0.041 [−0.006, +0.087]	0.086
	pooled external				+0.013 [−0.004, +0.031]	0.127
any impairment	CAUEEG (internal)	1098	0.753	0.749	−0.004 [−0.012, +0.003]	0.270
	ds004504	87	0.863	0.864	+0.002 [−0.023, +0.027]	0.889
	P-ADIC	116	0.596	0.595	−0.001 [−0.024, +0.023]	0.959
	BrainLat	80	0.637	0.658	+0.021 [−0.012, +0.054]	0.216
	MISP	48	0.641	0.606	−0.034 [−0.087, +0.018]	0.197
	pooled external				+0.002 [−0.013, +0.016]	0.812
MCI	CAUEEG (internal)	865	0.706	0.699	−0.007 [−0.016, +0.003]	0.192
	GENEEG	63	0.455	0.483	+0.028 [−0.081, +0.137]	0.620
	P-ADIC	74	0.521	0.522	+0.001 [−0.039, +0.041]	0.970
	MISP	48	0.603	0.588	−0.015 [−0.045, +0.016]	0.345
	pooled external				−0.007 [−0.031, +0.016]	0.544

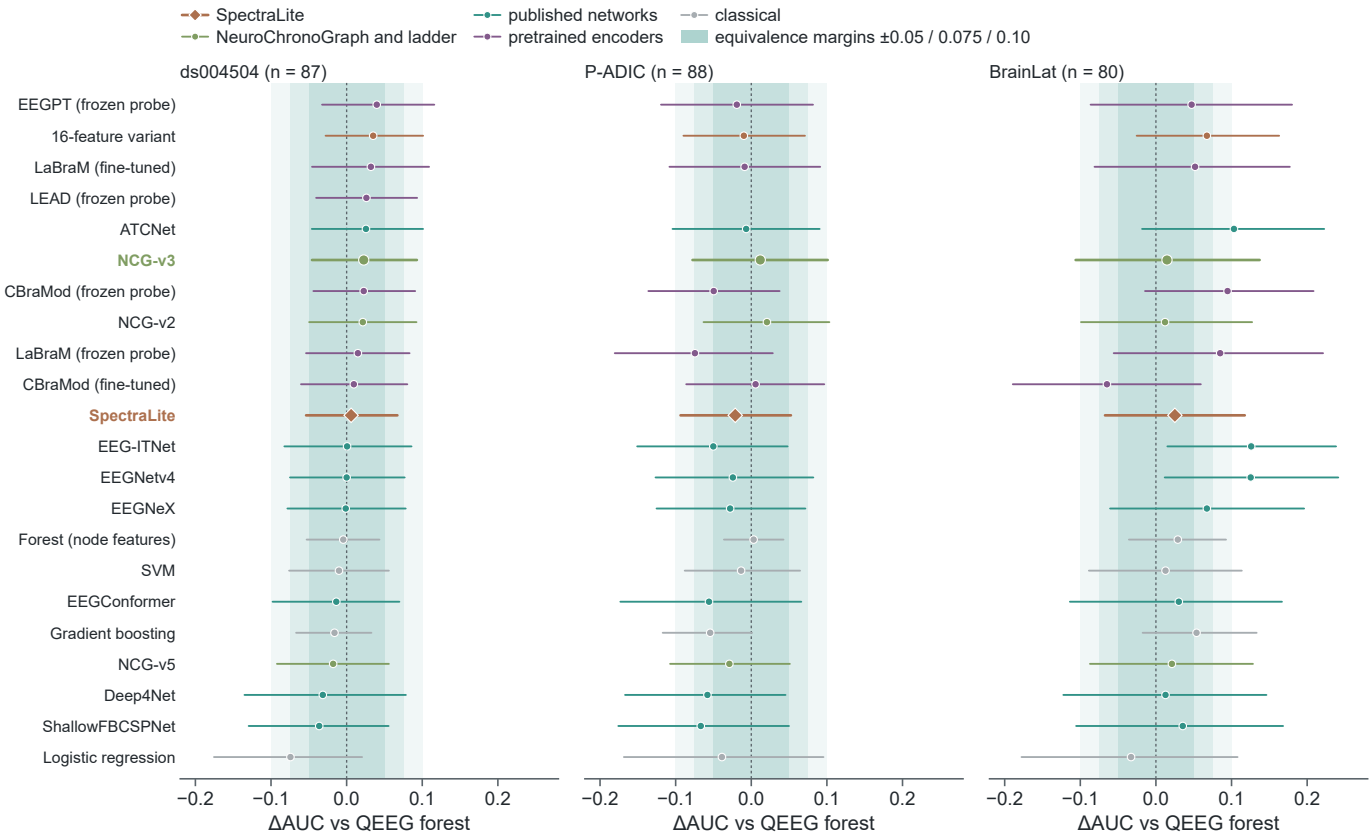


Fig. E2. Paired AUC difference from the QEEG forest (control versus dementia) at each external cohort: point, difference; line, 95% paired subject-bootstrap interval (4,000 resamples). Shaded bands mark the margins ±0.05 (prespecified), ±0.075, and ±0.10 (sensitivity analyses); equivalence requires the 90% interval inside a band. Colors give the model family. No difference survived Holm correction within any cohort.

identity passed by name. Frozen probes were standardized L_2 logistic heads on the subject-mean embedding; LaBraM and CBraMod were also fine-tuned end to end (10 epochs, AdamW, backbone 5×10^{-5}, head 10^{-3}, three seeds). Input scaling (microvolts divided by 100, or a per-window, per-channel z-score) was selected on internal out-of-fold AUC (Table E5); the selected and unselected scalings differed externally by up to 0.09 AUC, so the encoder results depend on a choice that used no external data but is partly arbitrary. The published pretraining corpora are motor-imagery, emotion, sleep, clinical (Temple University), and other non-dementia recordings for LaBraM [6], EEGPT [7], and CBraMod [8]. LEAD [9] was pretrained on 13 datasets including CAUEEG, P-ADIC, and BrainLat and fine-tuned by its authors on five others including ds004504; we used the pretraining-only checkpoint, so ds004504 and GENEEG are unexposed, while LEAD's CAUEEG out-of-fold AUC is not independent of pretraining and is excluded from internal comparisons. EEGPT's selected input is in physical units, so each BrainLat recording was first scaled to the CAUEEG median amplitude; without this label-free step its BrainLat dementia AUC was 0.42, and an unselected CBraMod probe on physical units reached 0.99 by detecting the export gain.

Precision. Table E6 gives the precision of the external comparisons. Sample sizes were fixed by the public releases; with 62–88 subjects per external task, a paired AUC difference had to exceed 0.085–0.136 to be detected with 80% power, and the probability of showing equivalence within the prespecified ±0.05 margin was below 2% even if two models were identical.

TABLE E5

FROZEN-PROBE AUC OF THE PRETRAINED ENCODERS UNDER BOTH INPUT SCALINGS. *SCALING SELECTED ON INTERNAL OUT-OF-FOLD AUC AND REPORTED IN THE BENCHMARK; THE INTERNAL VALUE OF THE SELECTED SCALING IS THEREFORE OPTIMISTIC. †LEAD WAS PRETRAINED ON CAUEEG, P-ADIC, AND BRAINLAT, SO ITS INTERNAL VALUE IS NOT INDEPENDENT OF PRETRAINING AND IT IS NOT SCORED AT P-ADIC OR BRAINLAT; DS004504 IS A LEAD FINE-TUNING DATASET, BUT THE PRETRAINING-ONLY CHECKPOINT USED HERE NEVER SAW IT.

Task	Encoder	Input scaling	CAUEEG	ds004504	P-ADIC	BrainLat
dementia	LaBraM	physical units	0.812	0.845	0.678	0.587
dementia	LaBraM	per-channel z-score*	0.819	0.855	0.630	0.673
dementia	EEGPT	physical units*	0.842	0.880	0.686	0.635
dementia	EEGPT	per-channel z-score	0.804	0.814	0.687	0.618
dementia	CBraMod	physical units	0.792	0.836	0.727	0.637
dementia	CBraMod	per-channel z-score*	0.798	0.863	0.655	0.683
dementia	LEAD	per-channel z-score*	0.853†	0.866	–	–
AD	LaBraM	physical units*	0.855	0.881	0.664	0.664
AD	LaBraM	per-channel z-score	0.836	0.840	0.637	0.720
AD	EEGPT	physical units*	0.861	0.874	0.671	0.697
AD	EEGPT	per-channel z-score	0.838	0.832	0.684	0.729
AD	CBraMod	physical units	0.810	0.853	0.696	0.691
AD	CBraMod	per-channel z-score*	0.824	0.883	0.662	0.745
AD	LEAD	per-channel z-score*	0.850†	0.881	–	–

TABLE E6

PRECISION OF THE EXTERNAL COMPARISONS. HALF-WIDTH, OF THE DELONG 95% INTERVAL OF THE MODEL'S AUC; SE, STRATIFIED PAIRED SUBJECT-BOOTSTRAP STANDARD ERROR OF ITS AUC DIFFERENCE FROM THE QEEG FOREST (4,000 RESAMPLES); MDD, MINIMUM DIFFERENCE DETECTABLE WITH 80% POWER AT TWO-SIDED $\alpha = 0.05$ ($2.80\times$SE); TOST POWER, THE PROBABILITY OF SHOWING EQUIVALENCE AT MARGINS 0.05 / 0.075 / 0.10 IF THE TRUE DIFFERENCE IS ZERO; n FOR 0.05, THE COHORT SIZE AT WHICH THE MDD WOULD BE 0.05, ASSUMING SE SCALES WITH $1/\sqrt{n}$.

Task	Cohort	Model	n	Half-width	SE	MDD	TOST power	n for 0.05
dementia	ds004504	SpectraLite	87	0.080	0.030	0.085	0.01 / 0.60 / 0.90	250
dementia	ds004504	16-feature variant	87	0.073	0.032	0.090	0.00 / 0.51 / 0.86	283
dementia	P-ADIC	SpectraLite	88	0.113	0.036	0.102	0.00 / 0.33 / 0.73	364
dementia	P-ADIC	16-feature variant	88	0.113	0.040	0.111	0.00 / 0.19 / 0.62	436
dementia	BrainLat	SpectraLite	80	0.126	0.048	0.133	0.00 / 0.00 / 0.35	567
dementia	BrainLat	16-feature variant	80	0.123	0.049	0.136	0.00 / 0.00 / 0.32	592
AD	ds004504	SpectraLite	65	0.074	0.030	0.085	0.00 / 0.59 / 0.90	188
AD	ds004504	16-feature variant	65	0.067	0.031	0.088	0.00 / 0.55 / 0.88	201
AD	P-ADIC	SpectraLite	88	0.112	0.037	0.104	0.00 / 0.30 / 0.71	379
AD	P-ADIC	16-feature variant	88	0.112	0.039	0.109	0.00 / 0.22 / 0.64	421
AD	BrainLat	SpectraLite	62	0.137	0.047	0.132	0.00 / 0.00 / 0.36	434
AD	BrainLat	16-feature variant	62	0.130	0.048	0.134	0.00 / 0.00 / 0.34	449

E4. COHORT SHIFT, AGE, EDUCATION, AND RECORDING CONDITION

Healthy controls differ between cohorts in most spectral features (Fig. E3a), and age differs between classes and cohorts (Fig. E3b). Repeating healthy-control cohort identification on age-matched control sets lowered the AUCs only slightly (Table E12). Supplementary Table S3 reports, for three tasks, the AUC of age alone, an age-adjusted AUC with a likelihood-ratio test of the score beyond age, and discrimination on age-matched pairs (BrainLat: 79 of 80 subjects have an age). At BrainLat, years of education separated controls from dementia with AUC 0.75 and from AD with 0.86; adjusted for age and education, SpectraLite scored 0.590 for dementia (score beyond both covariates $P = 0.071$) and 0.619 for AD ($P = 0.022$). Sex is released by ds004504, P-ADIC, and BrainLat; AUCs were mostly higher in women, with wide and overlapping intervals (Table E12).

CAUEEG recordings contain eyes-open, eyes-closed, and photic-stimulation segments, and the protocol uses the first 235 s regardless of condition. Labeling each analyzed window from the released event markers, 34% of windows were eyes-open in controls and in patients alike (AUC 0.51) and 1% overlapped photic stimulation; scoring only eyes-closed, photic-free windows changed SpectraLite's internal AUC from 0.855 to 0.845. Within CAUEEG the condition is therefore balanced between classes, but it differs from the eyes-closed external cohorts and may contribute to the cohort shift. GENEEG lacks T5 and T6; they were zero-filled before referencing, giving every fixed-montage network two identical synthetic channels. Scoring SpectraLite on the 17 real electrodes changed its MCI AUC by $+0.04$ to $+0.06$, both intervals including zero, and left it at chance. GENEEG was also recorded during an auditory oddball task.

Severity, recording condition, and eyes-closed training. Table E12 gives the AD severity strata (MMSE $\leq$20 versus $\geq$21

at ds004504 and P-ADIC; MoCA split at the case median at BrainLat), each scored against all controls with 2,000 subject-bootstrap intervals, and the AUC of P-ADIC after its AD cases were reweighted to ds004504's MMSE distribution (bins $\leq$15, 16–20, 21–24, $\geq$25). The any-impairment model, trained with CAUEEG's MCI cases as positives, did not detect P-ADIC's mild AD either (AUC 0.47–0.53 across six models). For the protocol analysis, the authors' preprocessed ds004504 (eyes closed) and ds006036 (photic session) files passed through the same pipeline as every cohort; 69 of 87 subjects had 24 clean windows in the photic session, which was eyes-open for a median 10% of the analyzed span. The CAUEEG threshold gave specificity 0.69 in the eyes-closed session and 0.62 in the photic session. Eyes-closed training kept 683 of 688 CAUEEG subjects (633 with all 24 windows, median 34 eyes-closed windows available) and used the canonical folds, seeds, and recipe.

MISP. MISP was analyzed after every other result in this paper. Before scoring, we wrote down that SpectraLite's control-versus-MCI AUC would lie between 0.40 and 0.65 with an interval including 0.5, that the median over classifiers would lie between 0.45 and 0.65, and that 20 local controls would restore specificity near 0.80 where the CAUEEG threshold would not; all three held (Table E12). All 61 recordings passed through the same pipeline and window rules as every cohort. The 17 control-versus-MCI models, plus the unselected input scaling of three frozen probes (20 classifiers), were applied unchanged; frozen-encoder probes were refitted on CAUEEG exactly as published (largest probability difference from the stored scores 7×10^{-4}), and LEAD is included because MISP is not in its pretraining data. Recording properties were compared between groups before any model was scored. MCI recordings were rejected more often (10 of 29 versus 3 of 32), mostly because every window of a recording exceeded 200 μV on some channel, and in set 2 the share of mains noise separated the groups (AUC 0.79). SpectraLite's AUC was 0.53 in set 1 and 0.71 in set 2. Scoring all 61 recordings, each rejected recording from its 24 least artifactual windows, gave 0.638 (0.54–0.73), with the gain confined to set 2 (set 1: 0.51), where mains noise also separated the groups, so we do not count it as MCI detection.

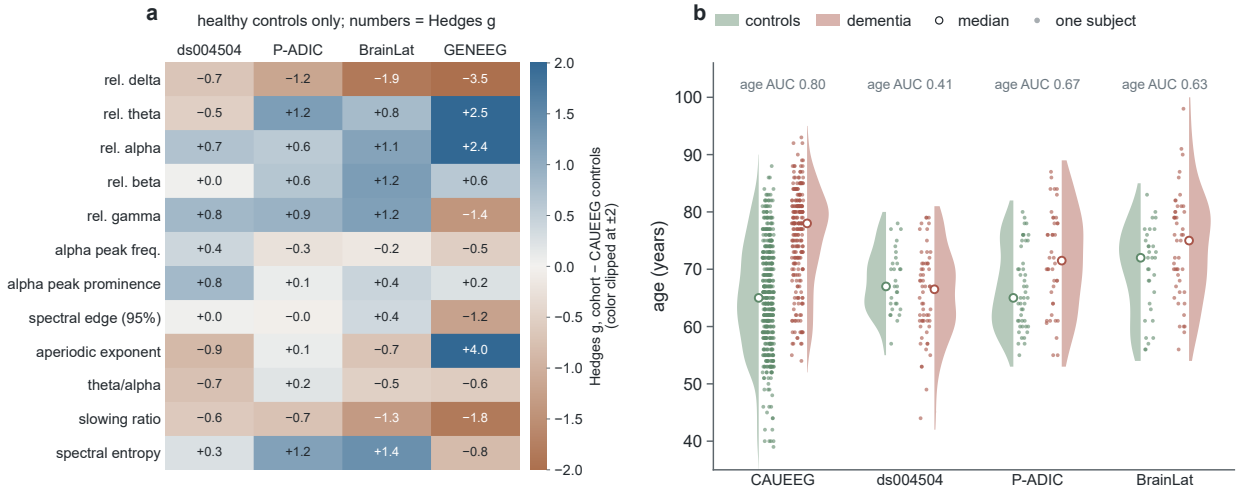


Fig. E3. Cohort shift. (a) Hedges g for each of the 12 base features, external-cohort healthy controls minus CAUEEG healthy controls; color is clipped at $\pm$2, numbers are unclipped. (b) Age by class in each cohort: half-violins are kernel densities, dots are subjects, open circles are medians; the number above each cohort is the AUC of age alone for control versus dementia (AD at P-ADIC).

E5. Montage, robustness, reliability, and attribution

Montage. Subsets of 8, 4, and 2 electrodes and two commodity four-electrode layouts (Muse-like T5/Fp1/Fp2/T6; Ganglion-like Fp1/Fp2/T3/T4) were cut from the cleaned 19-channel recordings and re-referenced to their own average, so this tests montage sensitivity; separately acquired headsets were not tested. SpectraLite keeps its 19-electrode weights and masks absent electrodes; a forest on the same 12 features, which cannot accept a different input size, is retrained for each subset (Fig. E4a). With 19 electrodes SpectraLite scored 0.850 at ds004504 (seed mean) and 0.79–0.82 on the subsets, against 0.79–0.83 for the forests; no paired difference survived Holm correction. An exploratory analysis of six four-electrode groups motivated by reports of temporal and temporo-parietal slowing [10], [11], run with the 16-feature variant, found no group better than all 19 electrodes (released results). These comparisons test whether a layout suffices for prediction; they do not locate the signal's sources.

Noise. Pink ($1/f$) or muscle-band (20–45 Hz) Gaussian noise of 5–20 μV RMS was added to every channel of the external recordings before feature extraction; neither model saw noise in training (Fig. E4b). Muscle-band noise of 10 μV RMS lowered the forest to 0.38 and SpectraLite to 0.69 (paired difference +0.30, 95% CI 0.11–0.48), whereas 20 μV pink noise favored the forest (0.814 against 0.755).

Reliability and cost. Each SpectraLite checkpoint occupied 302 KiB, or 92 KiB after dynamic INT8 quantization, which changed ds004504 dementia AUC from 0.841 to 0.840 (seed-42 ensemble). Scoring 24 cached feature windows took 0.63 ms

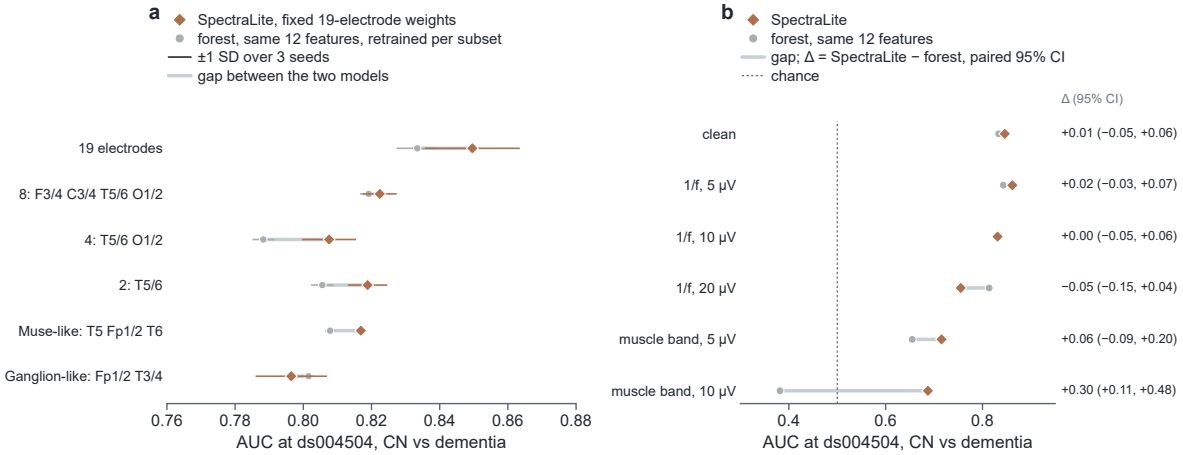


Fig. E4. Montage and robustness. (a) ds004504 control-versus-dementia AUC ($n = 87$) for electrode subsets: SpectraLite with fixed 19-electrode weights (diamonds) and a forest on the same 12 features, retrained for each subset (circles); lines, ± 1 SD over three training seeds. (b) The same AUC after adding pink or muscle-band noise to every channel (seed means); grey bar, the gap between the two models; dotted line, chance; Δ, paired difference SpectraLite minus forest with its 95% subject-bootstrap interval.

(FP32) or 0.95 ms (INT8) on one CPU thread, excluding preprocessing and feature extraction. AUC rose from 0.819 with one 4 s window to 0.841 with all 24 (Fig. E5a). Split-half reliability, ICC(2,1) between logits from the first and last 12 windows, was 0.96, and 14% of ds004504 referral decisions changed between halves at the source threshold (Fig. E5b).

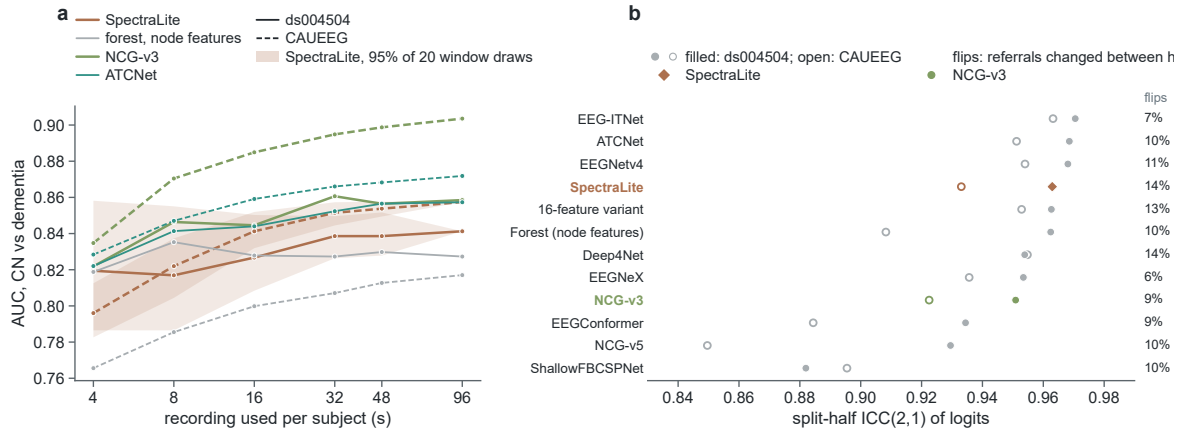


Fig. E5. Reliability. (a) Control-versus-dementia AUC when each subject is scored from K randomly drawn 4 s windows, mean of 20 draws, at ds004504 (solid) and CAUEEG out-of-fold (dashed); the band is the 2.5–97.5th percentile of the draws for SpectraLite. Seed-42 ensembles. (b) Split-half ICC(2,1) of subject logits; the column gives the percentage of ds004504 referral decisions that changed between halves.

Attribution. Integrated Gradients summarized which features the dementia model used across its 15 seed-fold checkpoints (Fig. E6); the ranking was stable across checkpoints (median pairwise Spearman ρ 0.89). The aperiodic exponent and the spectral edge ranked highest, as they did for the 16-feature variant. The ranking describes model behavior across a cohort; it does not explain single patients or locate cortical sources.

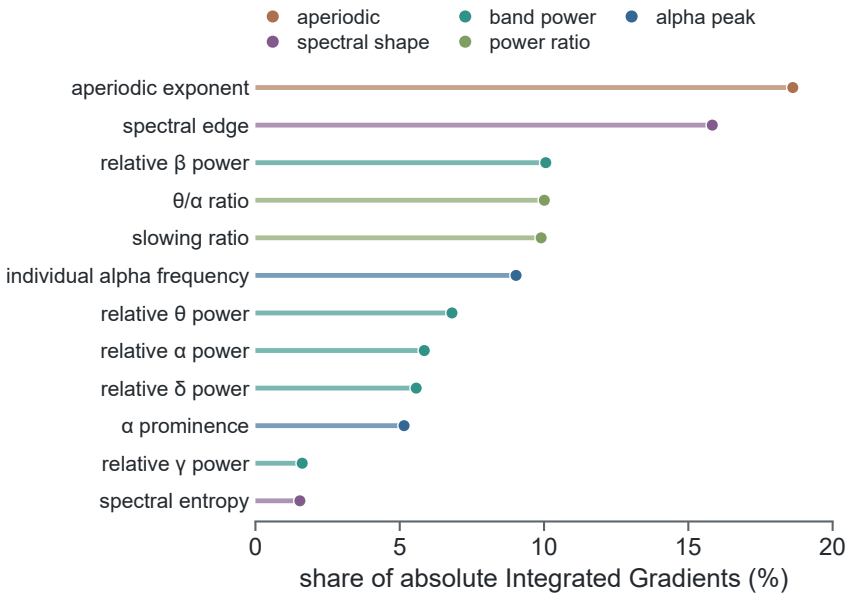


Fig. E6. Integrated Gradients (32 steps, mean CAUEEG-control baseline) for control versus dementia, SpectraLite: absolute attributions averaged over subjects and electrodes, as a share of the total, averaged over 15 seed-fold checkpoints.

E6. Operating points, calibration, and net benefit

Country and acquisition-stratum checks. Random splits within a cohort are exchangeable by construction, so the split-conformal rule must give mean specificity near $\lceil 0.8(k+1)\rceil/(k+1)$ there. As exploratory checks of that assumption, calibration controls were taken from one country (BrainLat) or one control acquisition stratum (P-ADIC) and evaluation subjects from the other (Table E7; triangles in main Fig. 4b,c). For SpectraLite and its variant, specificity departed from the target by up to 0.16 in both directions (the forest by up to 0.35), and for every model it stayed above the transferred threshold's. The strata are small and differ in diagnosis mix, every P-ADIC patient was recorded at 500 Hz, and sensitivity varied from 0.15 to 0.67, so these checks cannot estimate transfer to a new clinic.

Clinical metrics. Table E8 and Fig. E7a give operating characteristics at the split-conformal threshold for every model. Across every model, task, and external cohort, the mean held-out specificity of the conformal rule was 0.79–0.82. Because the external cohorts are case-enriched (38–67% cases), F1 and MCC overstate what a referral population would show; PPV and NPV at an assumed prevalence are the relevant summaries, and the NPV of referring no one at 10% prevalence is 0.90. Fig. E7b shows probability calibration, and Fig. E7c the net benefit of the two referral rules. Fig. E8 shows how sensitivity and predictive values change with the target specificity. The rule behaved the same for NCG-v3: at the 0.80 target its realized specificity was 0.80–0.81 and its sensitivity 0.67, 0.46, and 0.30 at ds004504, P-ADIC, and BrainLat, against SpectraLite's 0.70, 0.43, and 0.32.

TABLE E8: Clinical metrics for control versus dementia at every external cohort. Threshold-free: AUC, Brier score, expected calibration error (ECE, 10 equal-count bins). Operating point: split-conformal threshold from 20 local controls, mean over 200 splits (specificity on held-out controls, sensitivity on all cases); F1 and MCC at the cohort's own prevalence, PPV_{10}/NPV_{10} at 10% prevalence. The same metrics at the transferred threshold and for the other tasks are in clinical_metrics.json.

Model	AUC	Brier	ECE	Sens	Spec	BA	F1	MCC	PPV_{10}	NPV_{10}
ds004504, n = 87										
EEGPT (frozen probe)	0.880	0.162	0.155	0.75	0.80	0.77	0.81	0.52	0.30	0.97
16-feature variant	0.875	0.145	0.101	0.74	0.80	0.77	0.80	0.51	0.29	0.96
LaBraM (fine-tuned)	0.872	0.185	0.192	0.74	0.81	0.77	0.80	0.52	0.30	0.96
LEAD (frozen probe)	0.866	0.160	0.141	0.74	0.80	0.77	0.80	0.51	0.29	0.96
ATCNet	0.866	0.142	0.069	0.75	0.81	0.78	0.81	0.53	0.30	0.97
CBraMod (frozen probe)	0.863	0.146	0.078	0.76	0.81	0.78	0.82	0.54	0.31	0.97
NCG-v3	0.863	0.153	0.131	0.67	0.81	0.74	0.76	0.45	0.28	0.96
NCG-v2	0.861	0.150	0.121	0.67	0.81	0.74	0.76	0.45	0.28	0.96
NCG ladder +raw	0.858	0.153	0.107	0.70	0.80	0.75	0.78	0.47	0.28	0.96
NCG ladder +raw+graph	0.856	0.161	0.136	0.67	0.80	0.74	0.76	0.45	0.27	0.96

Model	AUC	Brier	ECE	Sens	Spec	BA	F1	MCC	PPV_{10}	NPV_{10}
LaBraM (frozen probe)	0.855	0.162	0.135	0.72	0.81	0.76	0.79	0.50	0.30	0.96
NCG ladder +raw+graph+lobe	0.851	0.158	0.131	0.65	0.79	0.72	0.74	0.42	0.26	0.95
CBraMod (fine-tuned)	0.850	0.171	0.155	0.65	0.81	0.73	0.74	0.43	0.27	0.95
NCG ladder +graph	0.848	0.161	0.112	0.71	0.80	0.75	0.78	0.48	0.28	0.96
SpectraLite	0.846	0.159	0.112	0.70	0.80	0.75	0.78	0.47	0.28	0.96
EEG-ITNet	0.841	0.151	0.090	0.64	0.81	0.72	0.74	0.42	0.27	0.95
EEGNetv4	0.840	0.152	0.081	0.71	0.81	0.76	0.79	0.50	0.30	0.96
Forest (QEEG-123)	0.840	0.184	0.199	0.65	0.80	0.73	0.74	0.42	0.26	0.95
EEGNeX	0.839	0.157	0.063	0.69	0.81	0.75	0.77	0.47	0.29	0.96
Forest (node features)	0.836	0.184	0.173	0.63	0.79	0.71	0.72	0.39	0.25	0.95
SVM	0.830	0.191	0.176	0.61	0.80	0.71	0.72	0.39	0.25	0.95
EEGConformer	0.826	0.161	0.090	0.68	0.82	0.75	0.77	0.47	0.30	0.96
Gradient boosting	0.824	0.212	0.209	0.59	0.81	0.70	0.70	0.39	0.26	0.95
NCG-v5	0.822	0.194	0.158	0.62	0.80	0.71	0.72	0.40	0.26	0.95
Deep4Net	0.809	0.182	0.138	0.62	0.81	0.71	0.72	0.40	0.26	0.95
ShallowFBCSPNet	0.804	0.168	0.088	0.62	0.81	0.71	0.72	0.40	0.27	0.95
Logistic regression	0.766	0.229	0.212	0.63	0.81	0.72	0.73	0.41	0.27	0.95
P-ADIC, n = 88										
NCG-v2	0.726	0.225	0.115	0.50	0.80	0.65	0.58	0.32	0.22	0.94
NCG-v3	0.717	0.230	0.127	0.46	0.80	0.63	0.55	0.28	0.20	0.93
CBraMod (fine-tuned)	0.711	0.230	0.135	0.49	0.80	0.65	0.57	0.31	0.21	0.93
Forest (node features)	0.708	0.214	0.146	0.54	0.80	0.67	0.61	0.35	0.23	0.94
Forest (QEEG-123)	0.705	0.218	0.128	0.54	0.80	0.67	0.62	0.36	0.23	0.94
ATCNet	0.698	0.279	0.221	0.46	0.80	0.63	0.55	0.28	0.20	0.93
NCG ladder +graph	0.698	0.232	0.137	0.48	0.81	0.64	0.57	0.31	0.22	0.93
LaBraM (fine-tuned)	0.696	0.244	0.167	0.48	0.80	0.64	0.57	0.30	0.21	0.93
16-feature variant	0.695	0.237	0.167	0.47	0.80	0.64	0.56	0.29	0.21	0.93
SVM	0.692	0.228	0.108	0.48	0.81	0.65	0.57	0.31	0.22	0.93
EEGPT (frozen probe)	0.686	0.248	0.187	0.47	0.81	0.64	0.56	0.30	0.21	0.93
NCG ladder +raw+graph	0.685	0.239	0.126	0.46	0.81	0.63	0.55	0.28	0.21	0.93
NCG ladder +raw+graph+lobe	0.684	0.240	0.145	0.46	0.80	0.63	0.55	0.28	0.21	0.93
SpectraLite	0.684	0.238	0.142	0.43	0.80	0.62	0.53	0.26	0.20	0.93
EEGNetv4	0.681	0.270	0.197	0.42	0.80	0.61	0.51	0.24	0.19	0.93
EEGNeX	0.677	0.275	0.195	0.40	0.81	0.60	0.50	0.23	0.19	0.92
NCG ladder +raw	0.677	0.243	0.172	0.44	0.80	0.62	0.53	0.26	0.20	0.93
NCG-v5	0.676	0.243	0.177	0.38	0.81	0.59	0.48	0.21	0.18	0.92
Logistic regression	0.666	0.312	0.285	0.41	0.80	0.61	0.51	0.23	0.19	0.92
CBraMod (frozen probe)	0.655	0.264	0.204	0.37	0.80	0.59	0.47	0.19	0.17	0.92
EEG-ITNet	0.655	0.290	0.226	0.40	0.80	0.60	0.50	0.23	0.19	0.92
Gradient boosting	0.651	0.262	0.198	0.46	0.80	0.63	0.55	0.28	0.21	0.93
EEGConformer	0.649	0.324	0.270	0.33	0.80	0.56	0.42	0.14	0.15	0.91
Deep4Net	0.647	0.364	0.347	0.31	0.81	0.56	0.41	0.13	0.15	0.91
ShallowFBCSPNet	0.638	0.308	0.251	0.36	0.81	0.58	0.46	0.19	0.17	0.92
LaBraM (frozen probe)	0.630	0.342	0.329	0.35	0.81	0.58	0.44	0.17	0.17	0.92
BrainLat, n = 80										
EEG-ITNet	0.714	0.221	0.145	0.46	0.81	0.64	0.58	0.28	0.22	0.93
EEGNetv4	0.714	0.220	0.152	0.46	0.81	0.63	0.58	0.28	0.21	0.93
ATCNet	0.691	0.252	0.218	0.43	0.81	0.62	0.55	0.25	0.20	0.93
CBraMod (frozen probe)	0.683	0.240	0.185	0.41	0.81	0.61	0.54	0.24	0.20	0.93
LaBraM (frozen probe)	0.673	0.447	0.469	0.42	0.81	0.62	0.55	0.24	0.20	0.93
EEGNeX	0.656	0.247	0.165	0.39	0.81	0.60	0.51	0.22	0.19	0.92
16-feature variant	0.656	0.231	0.103	0.34	0.81	0.57	0.46	0.16	0.16	0.92
Gradient boosting	0.642	0.255	0.163	0.43	0.81	0.62	0.55	0.24	0.20	0.93
LaBraM (fine-tuned)	0.640	0.301	0.265	0.32	0.81	0.56	0.44	0.14	0.16	0.91
EEGPT (frozen probe)	0.635	0.275	0.187	0.34	0.81	0.57	0.46	0.16	0.16	0.92

Model	AUC	Brier	ECE	Sens	Spec	BA	F1	MCC	PPV_{10}	NPV_{10}
NCG ladder +raw	0.633	0.253	0.153	0.32	0.81	0.56	0.44	0.14	0.16	0.91
NCG ladder +raw+graph	0.629	0.253	0.135	0.38	0.82	0.60	0.50	0.20	0.18	0.92
ShallowFBCSPNet	0.624	0.262	0.178	0.27	0.81	0.54	0.39	0.09	0.14	0.91
EEGConformer	0.618	0.286	0.258	0.26	0.81	0.54	0.38	0.09	0.14	0.91
Forest (node features)	0.617	0.244	0.142	0.40	0.81	0.60	0.52	0.22	0.19	0.92
NCG ladder +graph	0.614	0.244	0.122	0.35	0.81	0.58	0.47	0.17	0.17	0.92
SpectraLite	0.613	0.244	0.165	0.32	0.81	0.57	0.45	0.15	0.16	0.92
NCG ladder +raw+graph+lobe	0.611	0.253	0.152	0.38	0.81	0.59	0.50	0.20	0.18	0.92
NCG-v5	0.609	0.290	0.218	0.35	0.81	0.58	0.48	0.18	0.17	0.92
NCG-v3	0.603	0.299	0.241	0.30	0.81	0.55	0.42	0.12	0.15	0.91
Deep4Net	0.601	0.299	0.276	0.34	0.81	0.58	0.47	0.17	0.17	0.92
SVM	0.601	0.247	0.166	0.28	0.82	0.55	0.39	0.11	0.14	0.91
NCG-v2	0.600	0.288	0.203	0.32	0.81	0.57	0.45	0.14	0.16	0.91
Forest (QEEG-123)	0.588	0.247	0.122	0.29	0.81	0.55	0.41	0.11	0.14	0.91
Logistic regression	0.555	0.347	0.304	0.19	0.82	0.50	0.29	0.01	0.10	0.90
CBraMod (fine-tuned)	0.523	0.335	0.267	0.31	0.81	0.56	0.43	0.13	0.15	0.91

TABLE E7

EXPLORATORY COUNTRY AND ACQUISITION-STRATUM CHECKS OF SPLIT-CONFORMAL THRESHOLDS: CALIBRATION CONTROLS FROM ONE STRATUM, EVALUATION CONTROLS FROM THE OTHER. BRAINLAT: ARGENTINA (AR) AND CHILE (CL), FROM THE RELEASE'S SITE FIELD; CHILE CONTRIBUTED AD BUT NO FTD CASES, AND SIX FTD CASES WITHOUT A RECORDED COUNTRY (FIVE FROM A LATER RELEASE, ONE ABSENT FROM THE DEMOGRAPHICS FILE) ARE EXCLUDED, SO BRAINLAT ROWS USE 42 OF 48 DEMENTIA CASES. P-ADIC: CONTROL RECORDINGS RELEASED AT 500 HZ AND 200 HZ (CONTROLS AGED 55 AND OVER); EVERY P-ADIC PATIENT WAS RECORDED AT 500 HZ, SO P-ADIC SENSITIVITY USES ALL CASES AND DOES NOT TEST A CHANGE IN PATIENT ACQUISITION. k, CALIBRATION CONTROLS; n, EVALUATION CONTROLS/CASES; WILSON 95% INTERVALS; BY DIAGNOSIS, CASES DETECTED OF CASES. TRANSFERRED, THE CAUEEG THRESHOLD ON THE SAME EVALUATION CONTROLS; RANDOM, MEAN SPECIFICITY WHEN THE SAME NUMBER OF CALIBRATION CONTROLS IS DRAWN AT RANDOM FROM BOTH STRATA (EXCHANGEABLE). THE STRATA ARE SMALL AND DIFFER IN DIAGNOSIS MIX, SO THESE CHECKS DO NOT ESTIMATE TRANSFER TO A NEW CLINIC.

Task	Model	Cohort	Calibrate → test	k	n	Specificity	Sensitivity	By diagnosis	Transferred	Random
dementia	SpectraLite	BrainLat	AR → CL	19	13/15	0.77 (0.50–0.92)	0.60 (0.36–0.80)	AD 9/15	0.54	0.82
dementia	SpectraLite	BrainLat	CL → AR	13	19/27	0.95 (0.75–0.99)	0.15 (0.06–0.32)	AD 4/15; FTD 0/12	0.26	0.87
dementia	SpectraLite	P-ADIC	500 Hz → 200 Hz	35	11/42	0.73 (0.43–0.90)	0.45 (0.31–0.60)	–	0.27	0.81
dementia	SpectraLite	P-ADIC	200 Hz → 500 Hz	11	35/42	0.94 (0.81–0.98)	0.31 (0.19–0.46)	–	0.34	0.84
dementia	16-feature variant	BrainLat	AR → CL	19	13/15	0.77 (0.50–0.92)	0.60 (0.36–0.80)	AD 9/15	0.54	0.81
dementia	16-feature variant	BrainLat	CL → AR	13	19/27	0.89 (0.69–0.97)	0.19 (0.08–0.37)	AD 5/15; FTD 0/12	0.53	0.86
dementia	16-feature variant	P-ADIC	500 Hz → 200 Hz	35	11/42	0.64 (0.35–0.85)	0.57 (0.42–0.71)	–	0.36	0.81
dementia	16-feature variant	P-ADIC	200 Hz → 500 Hz	11	35/42	0.91 (0.78–0.97)	0.33 (0.21–0.48)	–	0.40	0.84
dementia	forest	BrainLat	AR → CL	19	13/15	0.54 (0.29–0.77)	0.67 (0.42–0.85)	AD 10/15	0.38	0.81
dementia	forest	BrainLat	CL → AR	13	19/27	0.84 (0.62–0.94)	0.15 (0.06–0.32)	AD 4/15; FTD 0/12	0.37	0.88
dementia	forest	P-ADIC	500 Hz → 200 Hz	35	11/42	0.45 (0.21–0.72)	0.60 (0.44–0.73)	–	0.36	0.82
dementia	forest	P-ADIC	200 Hz → 500 Hz	11	35/42	0.97 (0.85–0.99)	0.38 (0.25–0.53)	–	0.31	0.84
AD	SpectraLite	BrainLat	AR → CL	19	13/15	0.69 (0.42–0.87)	0.53 (0.30–0.75)	AD 8/15	0.46	0.82
AD	SpectraLite	BrainLat	CL → AR	13	19/15	0.95 (0.75–0.99)	0.27 (0.11–0.52)	AD 4/15	0.37	0.87
AD	SpectraLite	P-ADIC	500 Hz → 200 Hz	35	11/42	0.73 (0.43–0.90)	0.45 (0.31–0.60)	–	0.27	0.81
AD	SpectraLite	P-ADIC	200 Hz → 500 Hz	11	35/42	0.94 (0.81–0.98)	0.29 (0.17–0.44)	–	0.34	0.84
AD	16-feature variant	BrainLat	AR → CL	19	13/15	0.77 (0.50–0.92)	0.60 (0.36–0.80)	AD 9/15	0.54	0.82
AD	16-feature variant	BrainLat	CL → AR	13	19/15	0.89 (0.69–0.97)	0.27 (0.11–0.52)	AD 4/15	0.53	0.86
AD	16-feature variant	P-ADIC	500 Hz → 200 Hz	35	11/42	0.64 (0.35–0.85)	0.55 (0.40–0.69)	–	0.27	0.81
AD	16-feature variant	P-ADIC	200 Hz → 500 Hz	11	35/42	0.94 (0.81–0.98)	0.33 (0.21–0.48)	–	0.40	0.84
AD	forest	BrainLat	AR → CL	19	13/15	0.69 (0.42–0.87)	0.67 (0.42–0.85)	AD 10/15	0.38	0.80
AD	forest	BrainLat	CL → AR	13	19/15	0.84 (0.62–0.94)	0.27 (0.11–0.52)	AD 4/15	0.32	0.87
AD	forest	P-ADIC	500 Hz → 200 Hz	35	11/42	0.45 (0.21–0.72)	0.60 (0.44–0.73)	–	0.36	0.83
AD	forest	P-ADIC	200 Hz → 500 Hz	11	35/42	0.97 (0.85–0.99)	0.38 (0.25–0.53)	–	0.40	0.84

TABLE E9

SPECTRALITE OPERATING CHARACTERISTICS WITH 95% INTERVALS (2,000 RESAMPLES, EACH OF ONE CALIBRATION SPLIT WITH ITS HELD-OUT CONTROLS AND ALL CASES RESAMPLED; DERIVED METRICS FROM THE RESAMPLED SENSITIVITY AND SPECIFICITY). PPV AND NPV AT AN ASSUMED 10% PREVALENCE (5% AND 20% IN THE RELEASED RESULTS); NPV_{10} GAIN, NPV_{10} MINUS THE 0.90 OBTAINED BY REFERRING NO ONE; $NB_{0.10}$, NET BENEFIT AT 10% PREVALENCE AND THRESHOLD PROBABILITY 0.10, I.E. $0.10\,\mathrm{Sens} - 0.90\,(1-\mathrm{Spec})\,p_t/(1-p_t)$. THE 80% SPECIFICITY TARGET AND THESE PREVALENCES ARE ILLUSTRATIVE SCENARIOS, NOT RECOMMENDED OPERATING POINTS; FIG. E8 SHOWS OTHER TARGETS.

Cohort	Threshold	Sens	Spec	PPV_{10}	NPV_{10}	NPV_{10} gain	$NB_{0.10}$
Control versus dementia							
ds004504	transferred	0.76 (0.64–0.86)	0.68 (0.22–1.00)	0.21 (0.10–1.00)	0.96 (0.90–0.98)	+0.062 (+0.002–+0.081)	+0.044 (+0.001–+0.079)
	conformal	0.70 (0.59–0.83)	0.80 (0.33–1.00)	0.28 (0.11–1.00)	0.96 (0.92–0.98)	+0.060 (+0.016–+0.076)	+0.050 (+0.006–+0.078)
P-ADIC	transferred	0.81 (0.69–0.93)	0.33 (0.12–0.54)	0.12 (0.09–0.17)	0.94 (0.85–0.98)	+0.039 (−0.055–+0.078)	+0.014 (−0.010–+0.037)
	conformal	0.43 (0.24–0.67)	0.80 (0.50–1.00)	0.20 (0.10–1.00)	0.93 (0.90–0.95)	+0.028 (+0.002–+0.052)	+0.024 (+0.001–+0.045)
BrainLat	transferred	0.67 (0.54–0.79)	0.38 (0.08–0.75)	0.11 (0.07–0.21)	0.91 (0.67–0.96)	+0.012 (−0.233–+0.060)	+0.005 (−0.029–+0.042)
	conformal	0.32 (0.15–0.54)	0.81 (0.33–1.00)	0.16 (0.06–1.00)	0.92 (0.85–0.94)	+0.015 (−0.047–+0.036)	+0.013 (−0.019–+0.035)
Control versus AD							
ds004504	transferred	0.89 (0.78–0.97)	0.71 (0.33–1.00)	0.26 (0.13–1.00)	0.98 (0.95–1.00)	+0.083 (+0.054–+0.097)	+0.060 (+0.019–+0.092)
	conformal	0.78 (0.64–0.92)	0.80 (0.33–1.00)	0.30 (0.12–1.00)	0.97 (0.93–0.99)	+0.071 (+0.033–+0.088)	+0.058 (+0.017–+0.086)
P-ADIC	transferred	0.83 (0.71–0.93)	0.33 (0.12–0.54)	0.12 (0.09–0.17)	0.95 (0.86–0.98)	+0.047 (−0.038–+0.081)	+0.016 (−0.007–+0.040)
	conformal	0.43 (0.24–0.67)	0.80 (0.50–1.00)	0.20 (0.10–1.00)	0.93 (0.90–0.95)	+0.027 (+0.000–+0.050)	+0.023 (+0.000–+0.044)
BrainLat	transferred	0.70 (0.53–0.87)	0.41 (0.08–0.75)	0.12 (0.07–0.25)	0.93 (0.69–0.97)	+0.026 (−0.208–+0.070)	+0.011 (−0.025–+0.050)
	conformal	0.43 (0.23–0.63)	0.81 (0.42–1.00)	0.20 (0.08–1.00)	0.93 (0.88–0.95)	+0.028 (−0.025–+0.054)	+0.024 (−0.013–+0.053)

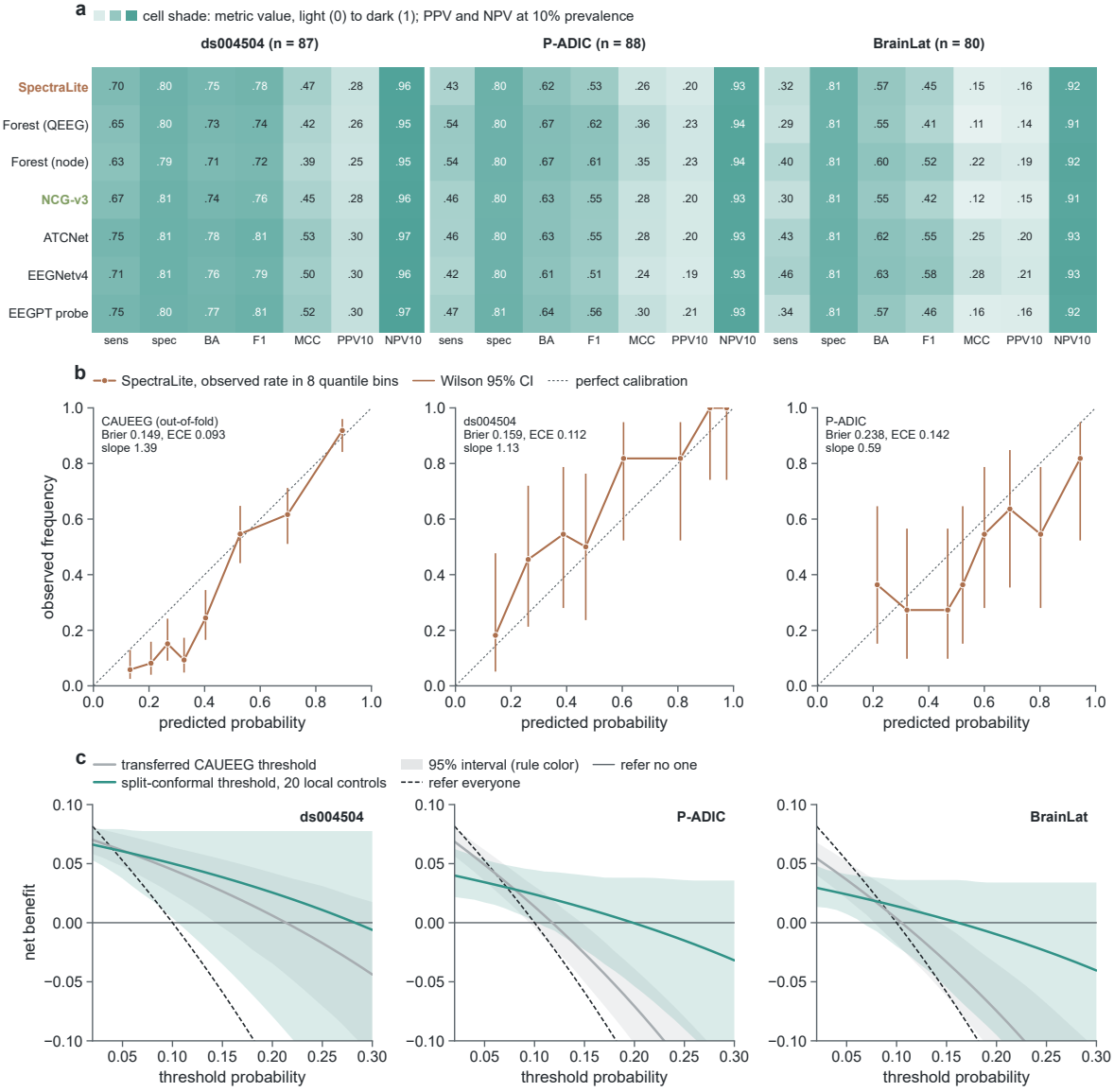


Fig. E7. Clinical metrics and calibration, control versus dementia. (a) Mean over 200 splits at the split-conformal threshold ($k = 20$ local controls): sensitivity, specificity, balanced accuracy (BA), F1 and Matthews correlation (MCC) at each cohort's own prevalence, and PPV and NPV at 10% prevalence; darker shading, higher value. (b) SpectraLite reliability diagrams at CAUEEG (out-of-fold), ds004504, and P-ADIC: observed dementia rate in eight quantile bins of predicted probability with Wilson 95% intervals; Brier score, expected calibration error (ECE, ten equal-count bins), and calibration slope in each panel. (c) Net benefit at an assumed 10% prevalence, $\pi\,\mathrm{sens} - (1-\pi)(1-\mathrm{spec})\,p_t/(1-p_t)$, against the threshold probability p_t, for the transferred CAUEEG threshold, the split-conformal threshold (mean over 200 splits), and referring everyone or no one. Bands are 95% intervals from 2,000 resamples, each drawing one calibration split and resampling its held-out controls and all cases.

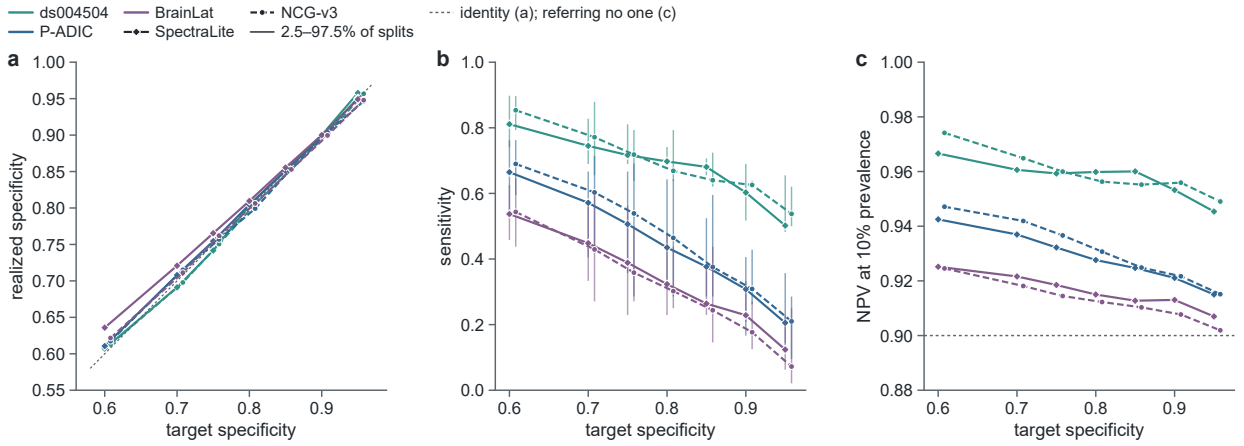


Fig. E8. Target specificity for the split-conformal rule (20 local controls, 200 random splits), control versus dementia, SpectraLite (solid, diamonds) and NCG-v3 (dashed, circles); the dependence on the number of local controls is in main Fig. 4d. (a) Realized mean specificity against the target; the diagonal is identity. (b) Mean sensitivity; vertical lines span the 2.5–97.5th percentiles of the splits. (c) Negative predictive value at an assumed 10% prevalence; the dashed line, 0.90, is the NPV of referring no one.

E7. Age normalization and local feature re-anchoring

Age normalization. For each of the 19×16 electrode-feature dimensions r, the controls of each CAUEEG training fold fit a quadratic age-dependent mean and a log-linear residual scale, $z_{ijr} = (x_{ijr} - \mu_r(a_i))/\sigma_r(a_i)$, where a_i is the subject's age, $\mu_r(a) = \phi_{r0} + \phi_{r1}a + \phi_{r2}a^2$, and $\log \sigma_r(a) = \omega_{r0} + \omega_{r1}a$; no external recording contributed. This lever was defined on the 16-feature variant; the normalized and raw models shared subjects, folds, recipe, and seeds. Age normalization raised internal AUC by 0.058–0.078, lowered external AUC at ds004504, raised it at P-ADIC, and left it unchanged at BrainLat (Table E10; Fig. E9); only control versus any impairment at ds004504 survived correction. With 20 local controls, split-conformal specificity stayed at 0.80–0.81 and sensitivity changed by -0.148 to $+0.131$ (lower at ds004504, higher at P-ADIC and BrainLat).

TABLE E10

Paired external AUC of the 16-feature variant with raw and age-normalized features. Δ is age-normalized minus raw with unadjusted stratified paired subject-bootstrap 95% intervals; p is the paired DeLong test, Holm-adjusted over the six prespecified ds004504 and P-ADIC comparisons. †BrainLat (79 of 80 subjects have an age) was not among the prespecified comparisons and is Holm-adjusted separately over its three tasks.

Contrast	Site	n	Raw AUC	Age-norm AUC	Δ (95% CI)	Holm p
CN vs dementia	ds004504	87	0.875	0.803	−0.072 [−0.139, −0.013]	0.117
CN vs dementia	P-ADIC	88	0.695	0.742	+0.047 [−0.035, +0.130]	0.780
CN vs dementia	BrainLat	79	0.652	0.633	−0.019 [−0.126, +0.082]	1.000†
CN vs AD	ds004504	65	0.909	0.863	−0.046 [−0.102, +0.001]	0.310
CN vs AD	P-ADIC	88	0.697	0.739	+0.041 [−0.040, +0.124]	0.780
CN vs AD	BrainLat	62	0.705	0.708	+0.003 [−0.095, +0.098]	1.000†
CN vs impaired	ds004504	87	0.864	0.770	−0.095 [−0.166, −0.029]	0.044
CN vs impaired	P-ADIC	116	0.595	0.645	+0.050 [−0.036, +0.138]	0.780
CN vs impaired	BrainLat	79	0.654	0.621	−0.033 [−0.150, +0.082]	1.000†

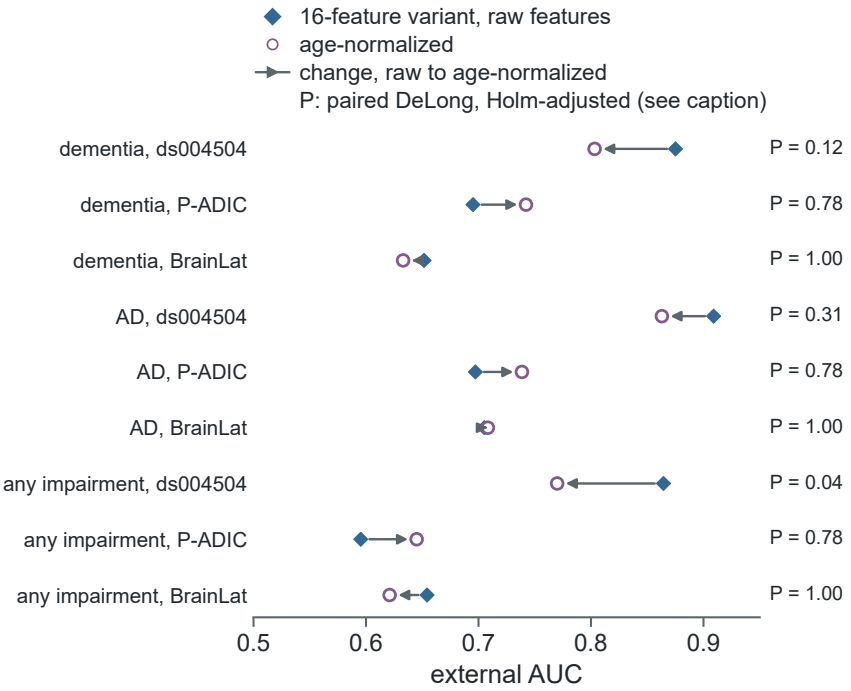


Fig. E9. External AUC of the 16-feature variant with raw and age-normalized features for three tasks at the three external cohorts with ages; arrows point from raw to normalized; P is the paired DeLong test, Holm-adjusted over the six prespecified ds004504 and P-ADIC comparisons and separately over the three BrainLat comparisons.

Local feature re-anchoring. At ds004504 and P-ADIC, ten healthy subjects estimated a per-feature median and MAD and ten others set the score threshold; held-out controls and cases entered neither step. Over 200 splits, re-centering (with or without MAD scaling) gave no Holm-corrected AUC improvement and lowered mean sensitivity in all six task-cohort comparisons.

E8. Prespecified transfer experiments

A protocol written before these runs (released as `IMPROVEMENT_PROTOCOL_2026-10-02.md`; not registered) fixed the candidates, selection rules, and tests. CAUEEG-only candidates used the benchmark's subjects, folds, seeds, and recipe:

feature-space domain randomization (per-recording offsets, scale jitter, electrode dropout), early stopping on a 15% inner-validation split, a 400-tree forest on the subject-median 19×16 features, and the mean of the 16-feature variant and that forest; all were built on the variant, as the protocol fixed. Multi-cohort models were trained on each CAUEEG fold plus the two other external cohorts, weighting each cohort, and each class within it, equally; the held-out cohort was never used for training, selection, or thresholds. For each cohort, the candidate with the highest mean AUC at the other two was chosen (nested leave-one-cohort-out). Three families of six paired DeLong tests were Holm-corrected separately (Table E11; Fig. E10); none survived. Choosing by internal AUC selected SpectraLite (12 features) for every task.

TABLE E11

PRESPECIFIED TRANSFER EXPERIMENTS (INTERNAL PROTOCOL WRITTEN BEFORE THE RUNS; NOT REGISTERED). EACH ROW COMPARES THE MODEL IN COLUMN 3 (AUC A) WITH ITS REFERENCE (AUC B: THE CAUEEG-ONLY 16-FEATURE VARIANT FOR H1, CAUEEG-ONLY FOREST FOR H2, SPECTRALITE FOR H3) ON THE SAME HELD-OUT SUBJECTS. MULTI-COHORT MODELS WERE TRAINED ON CAUEEG AND THE OTHER TWO EXTERNAL COHORTS, NEVER ON THE COHORT SCORED. Δ = A − B WITH A 4,000-RESAMPLE STRATIFIED BOOTSTRAP 95% INTERVAL; P IS THE PAIRED DELONG TEST, HOLM-ADJUSTED WITHIN EACH HYPOTHESIS (SIX TESTS).

Task	Cohort	Model A	AUC A	AUC B	Δ [95% CI]	P_{Holm}
H1: multi-cohort vs CAUEEG-only 16-feature variant						
dementia	ds004504	multi-cohort variant	0.870	0.875	−0.005 [−0.035, +0.024]	1.00
dementia	P-ADIC	multi-cohort variant	0.641	0.695	−0.054 [−0.116, −0.002]	0.36
dementia	BrainLat	multi-cohort variant	0.639	0.656	−0.017 [−0.120, +0.093]	1.00
AD	ds004504	multi-cohort variant	0.895	0.909	−0.014 [−0.061, +0.028]	1.00
AD	P-ADIC	multi-cohort variant	0.650	0.697	−0.047 [−0.104, +0.008]	0.49
AD	BrainLat	multi-cohort variant	0.711	0.705	+0.006 [−0.105, +0.126]	1.00
H2: multi-cohort vs CAUEEG-only forest						
dementia	ds004504	multi-cohort forest	0.859	0.832	+0.026 [−0.015, +0.071]	0.86
dementia	P-ADIC	multi-cohort forest	0.672	0.714	−0.041 [−0.102, +0.013]	0.78
dementia	BrainLat	multi-cohort forest	0.600	0.635	−0.035 [−0.130, +0.059]	1.00
AD	ds004504	multi-cohort forest	0.895	0.894	+0.001 [−0.055, +0.056]	1.00
AD	P-ADIC	multi-cohort forest	0.671	0.713	−0.041 [−0.100, +0.010]	0.75
AD	BrainLat	multi-cohort forest	0.662	0.692	−0.029 [−0.144, +0.084]	1.00
H3: lever chosen on the other two cohorts vs SpectraLite						
dementia	ds004504	SpectraLite + forest	0.866	0.846	+0.020 [−0.011, +0.055]	0.81
dementia	P-ADIC	16-feature variant	0.695	0.684	+0.011 [−0.025, +0.051]	1.00
dementia	BrainLat	SpectraLite + forest	0.650	0.613	+0.037 [−0.003, +0.080]	0.35
AD	ds004504	SpectraLite + forest	0.910	0.897	+0.013 [−0.010, +0.042]	0.82
AD	P-ADIC	+ inner-validation stopping	0.699	0.693	+0.006 [−0.034, +0.042]	1.00
AD	BrainLat	SpectraLite + forest	0.707	0.665	+0.043 [+0.002, +0.090]	0.26

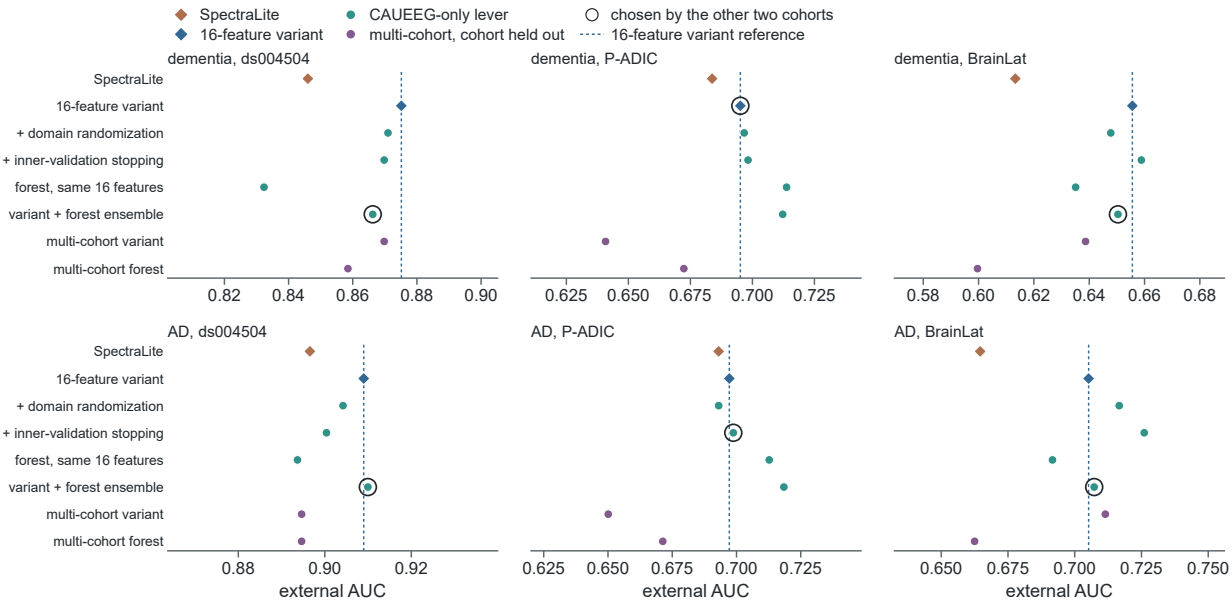


Fig. E10. External AUC of every prespecified candidate for control versus dementia (top) and AD (bottom), seed ensembles. The ring marks the candidate chosen for that cohort by its mean AUC at the other two cohorts; the dashed line is the 16-feature variant. Multi-cohort candidates (purple) were trained on CAUEEG and the two other external cohorts.

E9. Further checks

Table E12 collects further checks: bootstrap intervals and leave-family-out estimates for the capacity correlation; a decomposition of external AUC into cohort and model shares; the dementia model's separate AUCs for AD and FTD; healthy-control cohort identification after age matching; the transferred threshold recomputed per fold model on its own held-out controls; CAUEEG eye condition; GENEEG with T5 and T6 masked; education at BrainLat; and AUC by sex.

TABLE E12: Further checks. Capacity: Spearman ρ over the 11 networks trained from scratch (three-seed ensembles) with two-level bootstrap 95% intervals (networks and subjects, 2,000 resamples). Variance share: two-way decomposition of external AUC over the models scored at every external cohort, subject-bootstrap intervals. Cohort identity: logistic classifier on the QEEG vector, CAUEEG controls versus each cohort's controls after 1:1 matching within 3 years of age. Transferred specificity: CAUEEG threshold, all evaluation controls.

Check	Set	Result
Capacity ρ	internal	+0.69 (+0.01 to +0.98); without graph family +0.19 ($P = 0.65$)
Capacity ρ	ds004504	+0.23 (−0.74 to +0.89); without graph family −0.12 ($P = 0.78$)
Capacity ρ	P-ADIC	+0.40 (−0.56 to +0.92); without graph family −0.10 ($P = 0.82$)
Capacity ρ	BrainLat	−0.84 (−0.96 to +0.23); without graph family −0.71 ($P = 0.05$)
$\rho_{\text{int}} - \rho_{\text{ext}}$	ds004504	+0.46 (−0.35 to +1.50)
$\rho_{\text{int}} - \rho_{\text{ext}}$	P-ADIC	+0.29 (−0.40 to +1.21)
$\rho_{\text{int}} - \rho_{\text{ext}}$	BrainLat	+1.53 (+0.27 to +1.84)
Capacity ρ, 16-feature variant in place of SpectraLite	all sets	+0.66, +0.12, +0.40, −0.84
Variance share, dementia	26 models	cohort 0.89 (0.54–0.92); model 0.04 (0.02–0.20); residual 0.07
Variance share, AD	26 models	cohort 0.88 (0.46–0.92); model 0.05 (0.02–0.23); residual 0.08
Dementia model, AD vs FTD	ds004504, SpectraLite	AD 0.902 (0.832–0.972); FTD 0.754 (0.615–0.893, $n_{FTD} = 22$)
Dementia model, AD vs FTD	ds004504, 16-feature variant	AD 0.911 (0.844–0.978); FTD 0.817 (0.693–0.940, $n_{FTD} = 22$)
Dementia model, AD vs FTD	BrainLat, SpectraLite	AD 0.669 (0.532–0.806); FTD 0.521 (0.356–0.686, $n_{FTD} = 18$)
Dementia model, AD vs FTD	BrainLat, 16-feature variant	AD 0.705 (0.575–0.835); FTD 0.573 (0.412–0.733, $n_{FTD} = 18$)
Cohort identity, age-matched	ds004504	0.874 (0.766–0.965), 29 pairs; age alone 0.50
Cohort identity, age-matched	P-ADIC	0.975 (0.947–0.995), 79 pairs; age alone 0.49
Cohort identity, age-matched	BrainLat	0.936 (0.851–1.000), 32 pairs; age alone 0.50
Transferred specificity	ds004504, SpectraLite	fold ensemble, all controls 0.69; per-fold matched 0.69
Transferred specificity	P-ADIC, SpectraLite	fold ensemble, all controls 0.33; per-fold matched 0.37
Transferred specificity	BrainLat, SpectraLite	fold ensemble, all controls 0.38; per-fold matched 0.44
Transferred specificity	ds004504, 16-feature variant	fold ensemble, all controls 0.69; per-fold matched 0.70
Transferred specificity	P-ADIC, 16-feature variant	fold ensemble, all controls 0.39; per-fold matched 0.35
Transferred specificity	BrainLat, 16-feature variant	fold ensemble, all controls 0.53; per-fold matched 0.50
CAUEEG eyes-open windows	dementia	CN 0.34, case 0.34 (AUC 0.51); internal AUC SpectraLite 0.855→0.845, variant 0.836→0.828
CAUEEG eyes-open windows	MCI	CN 0.34, case 0.33 (AUC 0.49); internal AUC SpectraLite 0.703→0.698, variant 0.698→0.694
GENEEG, T5/T6 masked	SpectraLite	0.511 vs 0.455 zero-filled (Δ +0.056, −0.087 to +0.196)
GENEEG, T5/T6 masked	16-feature variant	0.527 vs 0.483 zero-filled (Δ +0.044, −0.063 to +0.154)
BrainLat education, dementia	SpectraLite	education alone 0.75; age- and education-adjusted AUC 0.590; P beyond both 0.071
BrainLat education, dementia	16-feature variant	education alone 0.75; age- and education-adjusted AUC 0.611; P beyond both 0.046
BrainLat education, AD	SpectraLite	education alone 0.86; age- and education-adjusted AUC 0.619; P beyond both 0.022
BrainLat education, AD	16-feature variant	education alone 0.86; age- and education-adjusted AUC 0.650; P beyond both 0.012
Sex, dementia	ds004504, SpectraLite	women 0.909 ($n = 44$), men 0.780 ($n = 43$); sex alone 0.59
Sex, dementia	ds004504, 16-feature variant	women 0.895 ($n = 44$), men 0.849 ($n = 43$); sex alone 0.59
Sex, dementia	P-ADIC, SpectraLite	women 0.711 ($n = 52$), men 0.609 ($n = 33$); sex alone 0.41
Sex, dementia	P-ADIC, 16-feature variant	women 0.695 ($n = 52$), men 0.647 ($n = 33$); sex alone 0.41
Sex, dementia	BrainLat, SpectraLite	women 0.585 ($n = 43$), men 0.590 ($n = 36$); sex alone 0.43
Sex, dementia	BrainLat, 16-feature variant	women 0.672 ($n = 43$), men 0.597 ($n = 36$); sex alone 0.43
AD severity strata	ds004504	SpectraLite mild 0.90 (0.78–0.99, $n = 8$), moderate 0.90 (0.82–0.97, $n = 28$); ρ with cognition −0.02; all models: mild 0.76–0.94, moderate 0.78–0.91
AD severity strata	P-ADIC	SpectraLite mild 0.52 (0.38–0.67, $n = 23$), moderate 0.88 (0.78–0.95, $n = 19$); ρ with cognition −0.62; all models: mild 0.44–0.60, moderate 0.74–0.91

Check	Set	Result
AD severity strata	BrainLat	SpectraLite mild 0.68 (0.49–0.86, $n = 12$), moderate 0.73 (0.53–0.89, $n = 13$); ρ with cognition −0.16; all models: mild 0.51–0.77, moderate 0.64–0.82
P-ADIC reweighted to ds004504 MMSE	P-ADIC	SpectraLite 0.684→0.779; all models +0.054 to +0.153
Mild AD, any-impairment model	P-ADIC	0.47–0.53 (6 models)
Same subjects, dementia	ds004504/ds006036 ($n = 69$)	preprocessing median +0.005 (−0.011 to +0.029); photic +0.015 (−0.028 to +0.070), 20 models; SpectraLite 0.847, 0.853, 0.832
Same subjects, AD	ds004504/ds006036 ($n = 55$)	preprocessing median +0.014 (−0.007 to +0.066); photic +0.034 (−0.013 to +0.077), 20 models; SpectraLite 0.873, 0.875, 0.894
Eyes-closed training, SpectraLite	CAUEEG ($n = 683$)	internal −0.015 (−0.028 to −0.002); ds004504 −0.017 (−0.055 to +0.019); P-ADIC −0.025 (−0.055 to +0.002); BrainLat +0.004 (−0.032 to +0.043)
Eyes-closed training, QEEG forest	CAUEEG ($n = 683$)	internal −0.012 (−0.029 to +0.007); ds004504 −0.015 (−0.049 to +0.015); P-ADIC −0.010 (−0.045 to +0.025); BrainLat −0.007 (−0.059 to +0.040)
MISP, MCI model	MISP ($n = 48$: 29/19)	SpectraLite 0.603 (0.47–0.73); set 1 0.53, set 2 0.71; forest 0.627 (0.50–0.75); 20 classifiers median 0.59 (0.44–0.72); none above the forest after Holm correction
MISP, other SpectraLite models	MISP	any-impairment model 0.641 (0.51–0.77); dementia model 0.642 (0.52–0.77)
MISP, all recordings	MISP ($n = 61$)	SpectraLite 0.638 (0.54–0.74); set 1 0.51, set 2 0.67; 13 classifiers median 0.61
MISP, recording properties	MISP	retained MCI 19/29, CN 29/32 (Fisher $P = 0.027$); line-noise share, set 2: 0.79 raw, 0.77 after cleaning; rejection rate 0.42; amplitude 0.46
MISP, thresholds (MCI model)	MISP	CAUEEG threshold specificity 0.45, sensitivity 0.74; 20 local controls specificity 0.81 (0.44–1.00), sensitivity 0.47
Variance share, MCI	GENEEG, P-ADIC, MISP	cohort 18% (1–61%), model 31% (12–45%); cohort means 0.53, 0.50, 0.57 (19 models)

E10. Spectral markers

Experiment 264 asks which of SpectraLite's 12 markers carry disease effects that replicate across cohorts, whether healthy baselines transfer, and whether the false referrals of a transferred threshold follow the markers. Its definitions and predictions were written on 2026-10-23, before the script existed, to `PREDICTIONS_264_MARKERS_2026-10-23.md` (in the GitHub repository). Three predictions restate exploratory observations from an earlier audit (C1–C3); eight are new (N1–N8). Nothing in it selects a model, feature, or threshold. Per subject, a marker is the median over the 24 windows, averaged over electrodes (whole head) or over a region. Effects are Hedges g with 2,000-resample subject-bootstrap CIs, pooled by DerSimonian–Laird random effects. The disease axis is an L_2 logistic regression ($C = 1$) on the 12 whole-head markers, standardized by CAUEEG controls and fitted on CAUEEG control versus dementia (out-of-fold AUC 0.78); an external control's axis value is its linear predictor. Table E13 lists every prediction; main Fig. 5 shows the disease effects, control shifts, and disease axis, and Fig. E11 the reliability and attribution estimates.

Seven of the eleven predictions held. Nine markers moved in the same direction with AD at all four dementia cohorts, and theta/alpha did so with little heterogeneity ($I^2 = 0.07$). Healthy baselines differed between cohorts: every marker had a cohort whose controls differed from CAUEEG controls by $|g| \geq 0.5$, and for every marker the largest shift exceeded the pooled disease effect. Among the 107 external controls, a control's position on the axis tracked its SpectraLite score within each cohort as well as pooled ($\rho = 0.60$–0.70 within cohorts), and the cohorts whose controls sat furthest along it lost the most specificity. This link is partly expected, because SpectraLite reads these markers; what it adds is that a transparent 12-number summary fitted on development data accounts for which healthy people a transferred threshold refers. The association with MMSE came from P-ADIC ($\rho = -0.54$ for theta/alpha); at ds004504 it was −0.07.

Four predictions failed. Theta/alpha rose slightly with MCI (pooled $g = 0.32$), which we had not expected; at that size it would give an AUC near 0.59, in line with the weak MCI detection. Posterior and frontal theta/alpha effects were nearly equal. A theta/alpha cutoff transferred from CAUEEG controls missed the 0.80 target at all three cohorts, but mostly upward (0.97, 0.76, 0.91), so the prediction of a loss at two cohorts failed; local cutoffs gave 0.79–0.82. Between the eyes-closed session and the photic session of the same 69 people, the markers hardly agreed (median ICC 0.04), and a subject's photic recording matched its own eyes-closed recording no better than chance (median rank 31 of 69), although each session separated AD from controls (theta/alpha $g = 1.39$ and 1.64). The photic session was recorded with eyes open under stepped 5–30 Hz stimulation, which may dominate each window's band powers; the release states that participant numbers match, and we cannot exclude a numbering error. The same eyes-closed recordings through our pipeline and through the authors' preprocessing agreed closely (ICC 0.95–1.00; descriptive, added after the predictions were fixed).

TABLE E13
FIXED PREDICTIONS OF THE MARKER ANALYSIS (EXPERIMENT 264) AND THEIR OUTCOMES. C, CONFIRMATORY RESTATEMENT OF AN EXPLORATORY OBSERVATION; N, NEW. EFFECTS ARE HEDGES g (95% CI) POOLED BY RANDOM EFFECTS OVER THE FOUR DEMENTIA OR THE FOUR MCI COHORTS; RELIABILITY IS SPEARMAN–BROWN SPLIT-HALF OR ICC(3,1). EVERY PREDICTION IS REPORTED.

	Prediction	Result	Outcome
C1	Pooled AD effect: θ, θ/α, slowing >0; IAF <0	θ/α 0.98 (0.82 to 1.15); slowing 0.70 (0.56 to 0.84); IAF -0.89 (-1.27 to -0.50)	held
C2	$\geq$6 markers with a control shift $\lvert g\rvert \geq 0.5$	12 of 12 markers	held
C3	No pooled MCI effect $\lvert g\rvert \geq 0.3$ with CI excluding 0	θ/α 0.32 (0.20 to 0.44)	not held
N1	Split-half reliability $\geq$0.80 at every cohort	median 0.96–0.98	held
N2	Eyes-closed vs photic ICC $\geq$0.60	median 0.04	not held
N3	Posterior $>$ frontal θ/α effect at $\geq$3 cohorts	1 of 4 (differences $\leq$0.06)	not held
N4	θ/α, slowing fall and IAF rises with MMSE (AD)	ρ -0.35, -0.35, 0.35	held
N5	Controls' axis value tracks score; separates false referrals	ρ 0.71 (0.58–0.80); AUC 0.85 (0.77–0.91)	held
N6	Cohort order of control shift = reverse of specificity	P-ADIC, BrainLat, ds004504 in both	held
N7	Transferred θ/α cutoff specificity $<$0.80 at $\geq$2 cohorts; local within 0.05	transferred 0.97, 0.76, 0.91; local 0.79, 0.82, 0.81	not held
N8	Attribution share rises with $\lvert$pooled AD $g\rvert$	ρ 0.55	held

Residual artifact burden (exploratory). Supplementary Section S4 and Table S4 give these unregistered sensitivity analyses (experiment 265), how they were run, and why they cannot show independence from muscle artifact. Muscle components were kept, and two of SpectraLite's markers (relative gamma power and the aperiodic exponent, fitted to 40 Hz) reach into the band where muscle activity dominates. The likelihood-ratio test compares an unpenalized logistic model of case status on the standardized artifact measures with the same model plus the standardized score (1 degree of freedom). The exclusion analysis leaves 37 cases at ds004504, 25 at P-ADIC, and 31 at BrainLat. Among controls, SpectraLite's score rose with the muscle proxy at ds004504 ($\rho = 0.38$) and BrainLat (0.30) but not at P-ADIC (0.02), so muscle activity may contribute to false referrals. GENEEG's rejection rate alone separated MCI from controls (AUC 0.79), and MISP lost 10 of 29 MCI but 3 of 32 control recordings to rejection, so both MCI results remain exploratory.

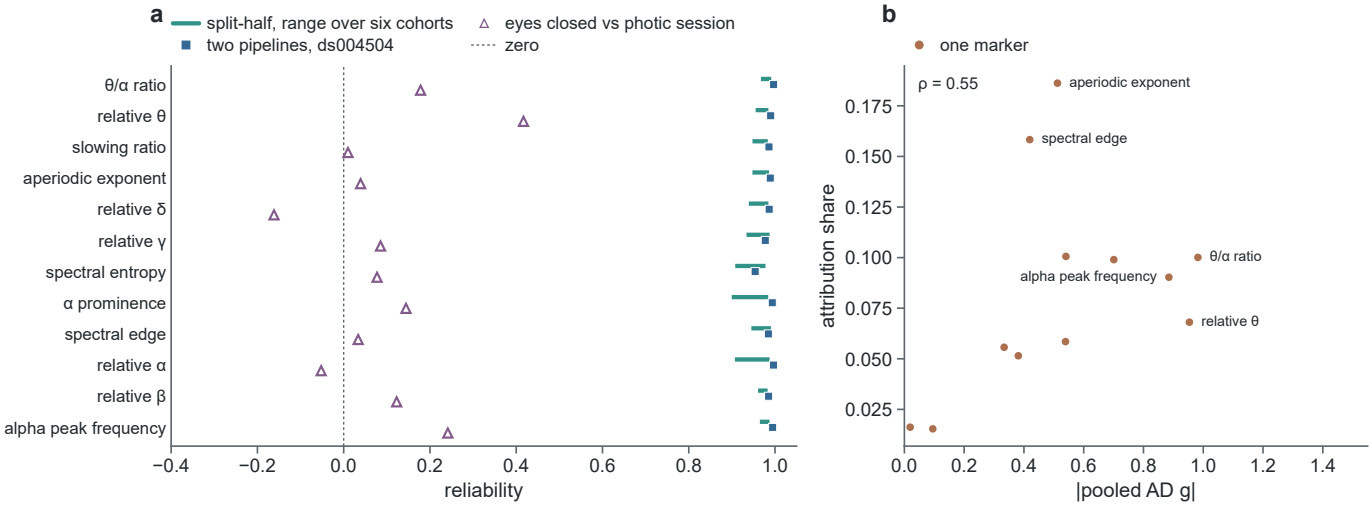


Fig. E11. Reliability of SpectraLite's 12 spectral markers and the model's reliance on them (experiment 264; disease effects, control shifts, and the disease axis are in main Fig. 5). (a) Spearman–Brown split-half reliability, range over six cohorts (teal bar); ICC(3,1) between our pipeline and the authors' preprocessing of the same ds004504 recordings (squares, $n = 87$); and ICC(3,1) between the eyes-closed and photic sessions of the same people (triangles, $n = 69$); dotted line, zero. Markers in the order of main Fig. 5a. (b) SpectraLite's integrated-gradients attribution share against the absolute pooled AD effect; ρ is Spearman's correlation over the 12 markers.

E11. Reproducibility

The TRIPOD+AI checklist [12] is in Section E12. *Code:* the GitHub repository named in the main text, at release tag `jbhi-submission-2026-10`; every script documents its inputs and outputs, and the experiment number given in each section identifies the script. *Data:* OpenNeuro ds004504 version 1.0.9 and ds006036 version 1.0.5; P-ADIC from Dryad (doi:10.5061/dryad.8gtht76pw); CAUEEG, BrainLat, GENEEG, and MISP as released by the sources cited in Table E1. *Pretrained encoders:* the braindecode checkpoints `labram-pretrained`, `eegpt-pretrained`, and `cbramod-pretrained`, and LEAD's pretraining-only checkpoint (LEADv2, P-Base). *Software:* Python 3.12.10, PyTorch 2.11 with CUDA 12.8, NumPy 2.4, SciPy 1.18, scikit-learn 1.9, MNE-Python 1.12, mne-icalabel 0.9, and braindecode 1.5.2 (versions of the final analysis environment). Training used one NVIDIA RTX 5070 Laptop GPU with deterministic cuDNN kernels, a fixed CUBLAS workspace, and fixed seeds; latencies were measured on one CPU thread of the same machine.

E12. TRIPOD+AI checklist

Locations refer to the main paper unless marked S (submitted supplementary material) or E (this document).

TABLE E14: TRIPOD+AI reporting checklist. The study develops one model (SpectraLite, fixed before external scoring) and evaluates it on new external data beside 26 other classifier configurations (27 in all), including an exploratory 16-feature variant chosen after external scoring; each configuration is scored on the tasks that each external cohort supports. TRIPOD+AI is reporting guidance; it does not appraise risk of bias.

Item	Topic	Where addressed
1	Title identifies development/evaluation, population, outcome	Title: external evaluation (study type), EEG (predictors), dementia (outcome), classifiers and their referral thresholds (what is evaluated), independent clinical cohorts (setting and population). The retrospective design is stated in the abstract's second sentence and the individual cohorts in Methods, Design and cohorts
2	Abstract	Abstract
3a	Healthcare context and rationale	Introduction, paragraphs 1–3
3b	Target population and intended use	Introduction, paragraph 1 (identifying patients for specialist assessment); Discussion, clinical interpretation (not supported for rule-out or referral use)
3c	Known health inequalities	Discussion, severity and recording condition (age, education, severity, and population differences between cohorts); E1 and Table E1 (cohorts from Korea, Greece, Israel, Argentina and Chile, and general practice)
4	Objectives	Introduction, final paragraph (primary and secondary objectives)
5a	Data sources	Methods, Design and cohorts; Table 1; Table E1
5b	Dates of data collection	Table E1 (as published by each source; P-ADIC 2011–2019; not stated by the other sources)
6a	Study setting	Methods, Design and cohorts; Table 1 (memory clinic, neurology department, routine clinical EEG, research network, general practice)

Table E14, continued

Item	Topic	Where addressed
6b	Eligibility criteria	Methods, Design and cohorts and Preprocessing; E1; Table E1 (source diagnostic criteria)
6c	Treatments received	Not released by any cohort; Table E1 caption; Discussion, Limitations
7	Data pre-processing and quality checking	Methods, Preprocessing and partial harmonization; E1; Fig. E1; E4 (eye condition, GENEEG electrodes)
8a	Outcome definition and time horizon	Methods, Design and cohorts (clinical diagnosis at recording; cross-sectional)
8b	Outcome assessor qualifications	Table E1 (neurologists or clinicians per source); Discussion, Limitations (no biomarker confirmation)
8c	Blinding of outcome assessment	Not reported by the source cohorts; Discussion, Limitations
9a	Choice of initial predictors	Methods, Spectral representation; Introduction (literature basis)
9b	Predictor definitions	Methods, Spectral representation; E2 (equations and constants)
9c	Predictor assessor qualifications	Not applicable: predictors are computed from the signal
10	Study size	Methods, Training and evaluation (sizes fixed by the public releases; precision and minimum detectable differences); Table E6
11	Missing data	Methods, Preprocessing (window quota; retention per class, Fig. E1); masked electrodes; E4 (GENEEG T5/T6); Supplementary Table S3 (one BrainLat subject without age)
12a	How data were used	Methods, Design and cohorts and Training and evaluation (development cohort only; external cohorts never used for fitting; the 16-feature variant chosen after external scoring)
12b	Predictor handling	Methods, Spectral representation (fixed constants; no fitted scalers)
12c	Model type and building steps	Methods, SpectraLite (development; the variant's selection) and Comparators (NeuroChronoGraph); E2; E3 (Table E4)
12d	Heterogeneity across sites	Results, Model and cohort (variance decomposition), Components of the cohort effect, and Spectral markers of the referred controls; Figs. 2, 3, and 5; Table E12
12e	Performance measures	Methods, Training and evaluation and Operating points (discrimination, calibration, operating characteristics, equivalence, net benefit)
12f	Model updating	Methods, Operating points (threshold only; no coefficient updating); E7 (input-space alternatives)
12g	How predictions are computed	Methods, SpectraLite (mean window probability; five-fold ensemble)
13	Class imbalance	Methods, Training and evaluation (class-weighted loss and forests)
14	Fairness	Results, What the cohort effect is made of; Supplementary Table S3 (age); Table E12 (education at BrainLat; sex at ds004504, P-ADIC, BrainLat); Discussion, Limitations
15	Model output	Methods, SpectraLite (subject probability) and Operating points (binary referral)
16	Differences between development and evaluation data	Results, Residual differences between cohorts; Table 1; Fig. 2b; Fig. E3; Table E12
17	Ethics approval	Methods, Design and cohorts (source approvals; secondary analysis of de-identified data); Table E1 (each source's approval); this secondary analysis is eligible for exemption from review under the ICMR National Ethical Guidelines (2017)
18a	Funding	Acknowledgment
18b	Conflicts of interest	Acknowledgment
18c	Protocol availability	No protocol for the benchmark; the transfer experiments (E8) followed an internal protocol fixed before they were run; the marker-analysis predictions are on GitHub
18d	Registration	Not registered (retrospective secondary analysis)
18e	Data availability	Data and code availability statement (all cohorts are public)
18f	Code availability	Data and code availability statement (code, splits, and analysis scripts on GitHub; weights on Hugging Face)
19	Patient and public involvement	None
20a	Participant flow	Methods, Design and cohorts; Table 1; Fig. E1; Supplementary Table S1
20b	Participant characteristics	Table 1 (age, electrodes, condition); Table E1 (age, sex, cognitive scores, education by class)
20c	Comparison of development and evaluation data	Results, Residual differences between cohorts; Fig. E3; Table E1
21	Number of participants in each analysis	Every table and figure caption states n
22	Full model details	Methods, SpectraLite; Fig. 1b (parameter budget); E2; released weights
23a	Performance with confidence intervals	Results; Table 2 (DeLong intervals for SpectraLite); Table 3 (bootstrap intervals); Table E9 (predictive values and net benefit with intervals); Table E3 (paired comparisons with SpectraLite); Fig. 3; Table E2
23b	Heterogeneity of performance	Results (internal vs external, per cohort, per task); Fig. 2; Supplementary Fig. S1; Table E12
24	Model updating results	Results, Operating points; Table 3; Figs. 3c and 4; Table E7 (exploratory country and acquisition-stratum checks)
25	Overall interpretation	Discussion

Table E14, continued

Item	Topic	Where addressed
26	Limitations	Discussion, Limitations
27a	Poor quality or unavailable input data	Methods, Preprocessing (rejection rule); E5 (masked electrodes, added noise); Fig. E4
27b	User interaction requirements	Methods, Operating points (local control recordings; threshold rule)
27c	Next steps	Discussion, Future work

REFERENCES

[1] M.-j. Kim, Y. C. Youn, and J. Paik, "Deep learning-based EEG analysis to classify normal, mild cognitive impairment, and dementia: Algorithms and dataset," *NeuroImage*, vol. 272, p. 120054, 2023.

[2] A. Miltiadous, K. D. Tzimourta, T. Afrantou *et al.*, "A dataset of scalp EEG recordings of Alzheimer's disease, frontotemporal dementia and healthy subjects from routine EEG," *Data*, vol. 8, no. 6, p. 95, 2023, openNeuro accession ds004504.

[3] O. Shor, A. Glik, A. Yaniv-Rosenfeld *et al.*, "EEG p-adic quantum potential accurately identifies depression, schizophrenia and cognitive decline," *PLOS ONE*, vol. 16, no. 8, p. e0255529, 2021, dryad doi:10.5061/dryad.8gtht76pw.

[4] P. Prado, V. Medel, R. Gonzalez-Gomez *et al.*, "The BrainLat project, a multimodal neuroimaging dataset of neurodegeneration from underrepresented backgrounds," *Scientific Data*, vol. 10, p. 889, 2023.

[5] A. Mishra, D. Joffe, S. S. Telidevara, D. S. Oakley, and A. Liu, "On the challenges of detecting MCI using EEG in the wild," arXiv:2501.17871, 2025, introduces the GENEEG cohort.

[6] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, "Large brain model for learning generic representations with tremendous EEG data in BCI," in *International Conference on Learning Representations (ICLR)*, 2024.

[7] G. Wang, W. Liu, Y. He, C. Xu, L. Ma, and H. Li, "EEGPT: Pretrained transformer for universal and reliable representation of EEG signals," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[8] J. Wang, S. Zhao, Z. Luo, Y. Zhou, H. Jiang, S. Li, T. Li, and G. Pan, "CBraMod: A criss-cross brain foundation model for EEG decoding," in *International Conference on Learning Representations (ICLR)*, 2025, arXiv:2412.07236.

[9] Y. Wang, N. Huang, N. Mammone, M. Cecchi, and X. Zhang, "LEAD: An EEG foundation model for Alzheimer's disease detection," *Transactions on Machine Learning Research*, 2026, arXiv:2502.01678v5.

[10] F. H. Duffy, G. B. McAnulty, and M. S. Albert, "Temporoparietal electrophysiological differences characterize patients with Alzheimer's disease: a split-half replication study," *Cerebral Cortex*, vol. 5, no. 3, pp. 215–221, 1995.

[11] L. Tomasello, L. Carlucci, A. Laganà, S. Galletta, C. V. Marinelli, M. Raffaele, and P. Zoccolotti, "Neuropsychological evaluation and quantitative EEG in patients with frontotemporal dementia, Alzheimer's disease, and mild cognitive impairment," *Brain Sciences*, vol. 13, no. 6, p. 930, 2023.

[12] G. S. Collins, K. G. M. Moons, P. Dhiman *et al.*, "TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, p. e078378, 2024.